WORKING PAPER · MEASUREMENT AUDIT

Dissecting LLM News Sentiment Signals

Scoring Noise, Identity Priors, Batch Effects, Price Echoes, and Coverage Selection in Short-Horizon Return Predictability

A measurement audit on 36 large U.S. equities over July–October 2026. The same headlines scored under four conditions—named, anonymized, repeated, and dictionary—then re-scored in a pre-specified interleaved session reveal a marginal baseline whose significance depends on the scoring draw, a reliable but stochastic instrument, a mild negative identity prior masked by session drift and batch effects, predictive content concentrated in price-describing headlines, and a sample selected on realized volatility.

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JEL: G10 · G12 · C52 · C55
Keywords: large language models · sentiment analysis · measurement validity · look-ahead bias

Working Paper — Audit and Re-examination Draft

Dissecting LLM News Sentiment Signals: Scoring Noise, Identity Priors, Batch Effects, Price Echoes, and Coverage Selection in Short-Horizon Return Predictability

A Measurement Audit on 36 U.S. Large-Cap Firms, July–October 2026

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Abstract. Do large language model (LLM) sentiment scores on firm-specific news predict stock returns because of what the text *means*, or because of what the scorer *brings to* the text —and to the batch? We audit a short-horizon predictability result of the kind now standard in the LLM-finance literature using 542 web-searched headlines on 36 large U.S. firms, 341 of which fall inside a 74-trading-day window (July–October 2026) and aggregate to 240 relevance-qualified firm-days. The identical headline set is scored under four conditions —with firm identity visible, with identity removed, in independent repeat calls, and with the Loughran and McDonald (2011) dictionary —and then re-scored in a pre-specified third session that interleaves four named and four anonymized rounds at the batch level and adds a batch-composition control. The audit yields five findings. (i) The baseline next-day predictability ($t = 2.10$) survives a wild-cluster bootstrap ($p = 0.030$) but not Benjamini–Hochberg correction over the declared family of seven comparisons ($q = 0.062$), weakens with the same-day return controlled ($t = 1.82$), and is *draw-dependent*: across ten re-scorings of the identical headlines the coefficient is stable ($[0.012, 0.016]$) but the t-statistic ranges over $[1.47, 2.12]$, so nominal significance depends on the day the pipeline scores. (ii) A nominal identity effect —named scores exceeding anonymized scores by 4.75 points ($t = 5.14$) —was first exposed as a session artifact (the identical named protocol drifted +5.79 points across a twelve-hour gap, $t = 7.03$) and then settled by the interleaved experiment: the identity prior is real, *negative*, and about three points (−2.82; all four round-level estimates negative, round-level $t = 6.8$; batch-call-clustered $t = 2.7$; V-excluded −3.13, clustered $t = 4.8$), while the drift itself decomposes into a +3.6-point *batch-composition* effect (pooled $z = 3.4$ across two sessions with inverted run order —the same headline scores differently depending on which other headlines share its batch, a channel not previously quantified, whose linear-anchoring part provably cannot move levels and whose asymmetry we estimate) and a non-monotonic endpoint residual of about ±1 point; the predictive coefficient is indistinguishable across conditions in all three designs (cross-session $p = 0.89$, same-session $p = 0.91$, interleaved $p = 0.62$), as is the dictionary horse race ($p = 0.73$). (iii) Scores are reliable but stochastic: the intra-class correlation is 0.919 [0.90–0.93] pooled across sessions and 0.949 within each sitting —reliability is stable even while levels drift —and only 29.9% of headlines receive the identical score in every call; the noise itself decomposes into a response-level mood shared by the 25 headlines of one API response (13% of variance) and idiosyncratic per-headline noise. (iv) Predictability concentrates in the 15% of headlines that contain language about already-realized price moves ($t = 1.80$ versus $t = 0.61$ for clean headlines), a split that survives an item-by-item audit of its own classifier (82.7% precision; strict-subset point estimate unchanged at 0.0475) and leave-one-out jackknifes with zero sign flips; the same-day loading is essentially unchanged by anonymization (3.33 vs. 3.11), locating this contamination in the *text*, not in the scorer's memory. (v) News coverage is selected on realized volatility (mean |return| of 215 vs. 154 bps on covered versus uncovered firm-days; firm-and-day fixed-effects test $t = 2.76$), so the estimation sample overweights high-attention events. Together, the results characterize the LLM sentiment signal in this setting as *weak, echo-prone, identity- and batch-sensitive, stochastic, and selected*: each of the five channels is individually small, but none was previously measured jointly, and the joint pattern shows why headline predictability statistics alone cannot certify measurement validity. The interleaved re-fielding design that repairs this paper's own initial confound is executed here end-to-end; we argue that anonymized, repeated, echo-audited, session-interleaved, and batch-frozen scoring should be the default protocol before LLM sentiment is deployed as a return predictor.

JEL classification: G10; G12; C52; C55.

Keywords: large language models; sentiment analysis; return predictability; measurement validity; look-ahead bias; Loughran–McDonald dictionary; replication.

Data and code availability: all headlines, all nine score sets (including the interleaved session), prices, scripts, and

analysis outputs are archived in the accompanying replication package; every number is generated by deterministic scripts from archived inputs. *Declarations:* this draft reports a pre-specified five-hypothesis test family evaluated through seven primary comparisons; the revision-stage confirmatory contrasts of the interleaved session (identity, batch composition, draw stability) are reported outside that family and labeled as such; null results are reported as null results.

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1 Introduction

Large language models have moved from curiosity to instrument in empirical finance. Lopez-Lira and Tang (2026), in the *Journal of Financial Economics*, score firm-specific news headlines with ChatGPT and show that the scores predict next-day stock returns and that this predictability is not matched by dictionary-based tone measures. The finding has been replicated, extended, and contested across models, markets, and windows (Nie et al., 2024), and it sits at the center of a rapidly growing literature that treats LLM output as a measurement layer for text-based variables. The premise of that literature is that the LLM *reads the text*. The premise is testable, and the tests matter: if the scores partly reflect the model's knowledge about the *firm* rather than the *headline*, or the *already-realized price move* described in retrospective headlines, or the *luck of a single stochastic call*, or the *selection of which firm-days carry news at all*, then predictability statistics confound semantics with contamination. Glasserman and Lin (2023) show for in-sample windows that company identity itself distorts GPT sentiment scores enough to change backtest performance; Lopez-Lira (2025) show that LLMs memorize economic and financial data inside their training window; Didisheim et al. (2025) show the memory is stronger for indices than for individual stocks; Chen et al. (2025) show that LLMs extrapolate recent returns when asked to forecast; and Herrera-Poyatos et al. (2025) document large cross-model and repeat-call variability in LLM sentiment classifications. What no study has done is measure these channels *jointly*, on the *same* headlines, against the *same* return windows, under a common inference standard. That is what this paper does.

Our vehicle is deliberately modest: a firm-level panel of 36 large U.S. equities over 74 trading days (July 1 to October 14, 2026), 542 firm-specific headlines collected through a documented web-search protocol, of which 341 are dated inside the window and 240 relevance-qualified firm-days survive aggregation. The window postdates the plausible training cutoff of the scoring model, which removes the most literal form of look-ahead bias —memorized realized outcomes (Lopez-Lira, 2025; Didisheim et al., 2025) —and therefore isolates the subtler channels that remain operative out of sample. The same 341 headlines are scored four ways: (i) the original protocol, with firm identity visible; (ii) an *anonymized* protocol in which company names and tickers are replaced by "the company"; (iii) three *independent repeat calls* of the original protocol; and (iv) the Loughran and McDonald (2011) dictionary. Each condition changes exactly one ingredient of the measurement. We then pre-specify a family of five hypotheses — baseline predictability, identity effects, scoring stability, price-echo contamination, and coverage selection —and evaluate all of them under two-way clustered inference, wild-cluster bootstrap, day-cluster bootstrap difference tests, and Benjamini–Hochberg control of the false discovery rate over the declared family.

Five results summarize the audit.

First, the baseline result is real but fragile —and its significance is draw-dependent. Next-day returns regress on named LLM sentiment with $t = 2.10$ ($p = 0.035$) under two-way clustered errors, and the wild-cluster bootstrap confirms the signal is not an artifact of a handful of clusters ($p = 0.030$). But the same statistic ranks last in the family once the Benjamini–Hochberg correction is applied ($q = 0.062$), the coefficient loses conventional significance when the same-day return —which is itself strongly correlated with sentiment ($t = 3.3$) —is added as a control ($t = 1.82$), and re-scoring the identical headlines ten times across three sessions moves the t-statistic between 1.47 and 2.12: the point estimate is stable, but whether a single-call pipeline would have reported "significant" depends on the day it scored. In a sample of 240 firm-days, the honest summary is "marginal evidence," not "significant predictability."

Second, what looked like an identity prior was a session artifact —and unpicking the artifact exposed two real level channels. Removing firm identifiers reduces the average score by 4.75 points (paired $t = 5.14$) —a level shift the audit initially read as the out-of-sample analog of the "distraction effect" of Glasserman and Lin (2023). A same-session benchmark added in revision overturned that reading: the named and anonymized rounds were fielded twelve hours apart, and the identical named protocol drifted +5.79 points between sittings ($t = 7.03$), dwarfing a same-session residual of roughly one point. The revision then ran the experiment its own diagnosis demanded —a third sitting with four named and four anonymized rounds interleaved at the batch level, pre-specified before data collection —and the design settled the question in both directions at once. The interleaved identity effect is −2.82 points (all four round-level estimates negative; round-level $t = 6.8$; batch-call-clustered $t = 2.7$; V-excluded −3.13 at $t = 4.8$): anonymized scores sit systematically *above* their named twins, a real but mild pessimism prior, opposite in sign to the published shift. And the drift itself decomposed: re-scoring the identical headlines inside the original 542-article batches (same session, same endpoint) moves levels +3.61 points, replicated at +3.50 in a fourth sitting with the run order inverted (pooled $z = 3.4$) —the "session drift" is mostly a *batch-composition* effect, a measurement channel we believe has not previously been quantified, with a further twist: random re-batching of the same corpus moves levels too, linear anchoring provably cannot, and the asymmetry we estimate can. What identity does *not* do is equally clear and now confirmed under the definitive design: the return-predictive coefficient is statistically indistinguishable between the named and anonymized conditions whether the comparison is cross-session ($p = 0.89$), within session ($p = 0.91$), or interleaved with replication ($p = 0.62$), and the dictionary horse race is likewise indistinguishable ($p = 0.73$). The lesson is uncomfortable on both sides of the LLM-versus-dictionary debate: in a sample this size, the difference between instruments is not estimable, and any claim of dominance based on one instrument being significant while the other is not is a statistical error we document explicitly below.

Third, the instrument is stochastic. Four independent calls of the *same* prompt on the *same* headlines produce an intra-class correlation of 0.919 (95% CI $[0.90, 0.93]$) and only a 29.9% exact-replication rate across all four calls (44% for a single pair of calls); the within-headline standard deviation across calls is 7.8 points. This contradicts the intuition —stated in earlier drafts of this very project —that default decoding makes the instrument deterministic. Every downstream regression inherits this noise as attenuation.

Fourth, the predictive content sits where the text echoes prices. A text audit flags 15.2% of headlines as containing language about realized price moves ("rose 14%," "dropped by 5% after earnings"). The next-day coefficient on sentiment is 0.047 ($t = 1.80$) for firm-days with such headlines and 0.003 ($t = 0.61$) for firm-days without them. The anonymized condition does essentially nothing to the same-day loading of sentiment on returns ($t = 3.33$ named vs. 3.11 anonymized), which rules out the scorer's memory as the source: the echo is *in the text*. Because our window postdates the training cutoff, this channel is precisely the one the literature cannot dismiss as memorization —search-collected news archives systematically over-represent retrospective, price-descriptive coverage.

Fifth, the sample itself is selected. Covered firm-days exhibit mean absolute same-day returns of 215 bps against 154 bps for uncovered firm-days (firm-and-day fixed-effects test $t = 2.76$; see Section 5.5 for why the pooled two-sample statistic overstates this), and firm-level news counts correlate 0.48 with firm return volatility. The estimation sample is therefore an over-weight of high-attention, high-absolute-return events —exactly the environment in which short-horizon continuation and reversal are strongest for reasons unrelated to news semantics (Chan, 2003).

None of these channels is large enough, alone, to overturn a well-powered result. Jointly, they change what a small-sample LLM predictability statistic *means*. The contribution of this paper is therefore not a new anomaly but a *measurement audit protocol*: anonymization, repeat scoring, text-level echo flags, and coverage-selection diagnostics, all run on one frozen corpus under one inference standard —plus, in revision, the interleaved re-fielding design that repairs the confounds the first pass created. We release the full pipeline —headlines, all nine score sets, prompts, scripts, and every table and figure —as a replication

package, and we report the family-wise error control explicitly so that future studies can register audit families before examining returns.

The paper also carries a methodological signature we make no attempt to hide: the research pipeline, including the scoring instrument itself, was operated end-to-end with AI assistance. We treat that as an argument *for* the audit layer, not a caveat to be buried: when the marginal cost of producing an empirical result approaches zero, the binding constraint on scientific quality shifts from execution to verification. Every quantity in this paper was recomputed from archived artifacts in producing this draft, and the replication package is the source of truth.

The remainder proceeds as follows. Section 2 positions the audit against the LLM-finance and textual-analysis literature. Section 3 sets out the measurement decomposition that organizes the tests. Section 4 documents the corpus, the four scoring conditions, and the econometric design. Section 5 presents the five audit results. Section 6 reports robustness and the threats to validity we could and could not address. Section 7 discusses what survives, for whom, and under which protocol. Section 8 states limitations; Section 9 concludes.

2 Related Literature and the Research Gap

2.1 News text and stock returns

That quantified news text contains information about subsequent returns is one of the longest-standing results in empirical finance. Tetlock (2007) measures the pessimism of a daily press column and finds transient market-level pressure and reversal; Garcia (2013) shows news sentiment predicts market returns in recessions but not expansions; Chan (2003) documents that stocks with salient news drift and then reverse while no-news stocks mean-revert; and Da et al. (2011) show that attention itself, measured by search volume, predicts short-run pressure and reversal. The measurement instrument in this first generation is a dictionary —originally the Harvard psychological lists, then the finance-specific Loughran and McDonald (2011) lists that now dominate the field, and whose construction and properties are reviewed in Loughran and McDonald (2016). Dictionary measurement is transparent and cheap but blind to negation, context, and event semantics, a limitation already visible in the higher information content of supervised alternatives (Jegadeesh and Wu, 2013).

2.2 LLMs as measurement instruments

The current generation replaces the dictionary with an instruction-following LLM. Lopez-Lira and Tang (2026) is the reference point: ChatGPT scores of firm-specific headlines predict next-day returns and beat dictionary and FinBERT-style baselines on the same text —in the published *Journal of Financial Economics* version, 159,137 firm-headline observations on 4,123 U.S. stocks over October 2021–May 2024, daily hit rates near 90% for the initial reaction, a 2.97 Sharpe for the drift strategy, and a Sharpe that decays from 6.54 to 1.22 across the sample as LLM adoption spreads —the market-efficiency arc our audit's fragility findings rhyme with at small scale. The published version anchors a fast-growing literature surveyed by Nie et al. (2024), including zero-shot LLM sentiment integrated with price prediction (Officioso et al., 2026), cross-model comparisons (Teles et al., 2025), and prompt-engineered GPT-4o versus FinBERT classification (Kang et al., 2025). The premise of all of this work is that the LLM's score is a *function of the text*. The present paper interrogates that premise directly.

2.3 The audit strand: bias, memory, and stability

A newer strand asks what else the score is a function of. Glasserman and Lin (2023) are the direct precedent: they compare trading strategies built on original versus *de-identified* headlines inside the model's training window and find, surprisingly, that anonymized headlines outperform —company identity acts as a "distraction" that corrupts sentiment measurement, an effect they show is strongest for large firms.

Lopez-Lira (2025) demonstrate that LLMs recall pre-cutoff macroeconomic and financial data with near-lookup accuracy, that instructions and masking fail to suppress recall, and that post-cutoff there is no recall —which makes post-cutoff windows the clean environment for studying the *remaining* channels. Didisheim et al. (2025) show predictable memory in financial analysis, stronger at the index level than the individual-stock level, suggesting firm-level scores are less contaminated by recall. Chen et al. (2025) find that LLMs asked to forecast from historical returns extrapolate recent performance —a mechanism that matters here because web-searched corpora over-represent headlines that *describe* recent performance. On the instrument side, Herrera-Poyatos et al. (2025) document the "model variability problem": sentiment classifications differ materially across models and across repeated calls of the same model. And standardized look-ahead audits are beginning to appear as benchmarks —most recently Benhenda (2026)'s Look-Ahead-Bench, which scores point-in-time versus standard LLMs by alpha decay across market regimes —reflecting a field-level recognition that output-based evaluation cannot certify measurement validity. What none of these audit instruments measures is the scoring *context* itself: the batch the headline is scored in, the sitting it is scored in, and the identity it is scored with. The present audit adds those to the checklist.

2.4 The gap

Three gaps remain. First, Glasserman and Lin (2023)'s anonymization experiment is run *in-sample*; whether identity effects survive in a post-cutoff window —where they can no longer be memory —has not been tested, and the answer determines how much of the identity effect is a pure prior (about the firm) versus recall (about the outcome). Second, the contamination channels have been studied one at a time; no study measures identity, stochasticity, text-level price echoes, and sample selection *jointly* against the same return windows with formal difference tests and family-wise error control. Third, small-sample LLM predictability studies —including earlier drafts of this project —routinely compare instruments by the significance of one coefficient against another, a practice the econometrics literature rejects (e.g., Romano and Wolf, 2005) and that we correct here with bootstrap difference tests. The contribution of this paper is to fill these three gaps on one frozen corpus, and to convert the resulting checklist into a reusable protocol.

2.5 Hypotheses

The audit family, pre-specified before the augmented inference was run:

H1 (Baseline). Named LLM sentiment positively predicts the next-day close-to-close return, conditional on lagged return, lagged volatility, and the market factor, under two-way clustered inference, a wild-cluster bootstrap, and Benjamini–Hochberg control across the declared family.

H2 (Identity). Anonymizing firm identifiers (a) shifts the level of the score distribution and (b) leaves the predictive coefficient statistically unchanged, the two statements together implying an identity *prior* without identity *information*; the named-versus-dictionary difference is likewise tested directly rather than inferred from separate significances. [Revision note: part (a) as originally fielded is confounded with session drift; the pre-specified interleaved session of Section 5.2 resolves it —the level shift is -2.82 points (round-level $t = 6.8$; call-clustered $t = 2.7$), a genuine negative prior —and part (b) is robust in all three designs.]

H3 (Stability). Repeat calls of the identical prompt are highly but imperfectly correlated; the intra-class correlation quantifies the stochastic component of the instrument and the attenuation it implies.

H4 (Echo). Headlines containing realized-price language carry the predictive loading; clean headlines do not; and the same-day loading survives anonymization, locating the channel in the text.

H5 (Selection). Coverage (which firm-days carry news) is positively selected on realized absolute returns and volatility.

H2 and H4 are the hypotheses whose null results would have been hardest to publish a decade ago and are the most informative now: H2 distinguishes priors from information, and H4 distinguishes text semantics from text echoes of prices.

3 A Measurement-Decomposition Framework

The role of theory in this paper is narrow and deliberately so. We do not model price formation or claim to identify a mechanism (attention, liquidity provision, or otherwise); the earlier draft of this project interpreted an underreaction–overshooting–reversal pattern through a limited-attention model, and the audit below shows that the empirical signature feeding that interpretation —a large same-day loading of returns on sentiment —is substantially a property of the *measured text*, not of the price process. What is needed instead is an accounting framework that separates the ingredients of the score so that each can be tested. This section provides it in three equations; every empirical section then maps one-to-one onto a term.

3.1 The score equation

Let $S^N_{i,t}$ denote the named-condition sentiment score for firm i's headline at date t, and write

$$S^N_{i,t} = \underbrace{\kappa\,\theta_{i,t}}_{\text{semantic tone}} + \underbrace{\iota\,G_i}_{\text{identity prior}} + \underbrace{\rho\,P_{i,t}}_{\text{price echo}} + \varepsilon_{i,t}, \tag{1}$$

where $\theta_{i,t}$ is the news-relevant fundamental signal in the text, G_i is the scorer's firm-specific prior (any stable belief about firm i that is independent of the headline at hand), $P_{i,t}$ is the realized price movement *described by* the text (a retrospective headline such as "rose 14% in 30 days" has $P_{i,t} \neq 0$ by construction), and $\varepsilon_{i,t}$ is call-level scoring noise. The anonymized score $S^A_{i,t}$ is generated by the same equation with the identity channel switched off ($\iota = 0$ in effect, since the input no longer identifies the firm), and repeat calls identify $\mathrm{Var}(\varepsilon)$ directly. The dictionary score is a separate, low-duty-cycle instrument applied to the same text.

Equation (1) is an *accounting* device, not a behavioral model: it imposes no orthogonality among the components. Indeed, the interesting case is the non-orthogonal one, in which news that matters for θ is also more likely to be retrospectively described and more likely to be covered at all —the selection problem formalized in Section 3.3.

3.2 What each audit test identifies

The four scoring conditions deliver direct contrasts on the terms of (1):

$$S^N_{i,t} - S^A_{i,t} = \iota\,G_i + (\varepsilon^N_{i,t} - \varepsilon^A_{i,t}) \qquad \Rightarrow \text{H2: level shift and agreement } (\iota), \tag{2}$$

$$S^N_{i,t}(r) - S^N_{i,t}(r') = \varepsilon_{i,t}(r) - \varepsilon_{i,t}(r') \qquad \Rightarrow \text{H3: repeat-call variance}, \tag{3}$$

and the echo channel is tested by splitting the sample on a text-level flag rather than by a contrast of scores: if headlines that describe realized prices (the H4 flag) carry predictive loading that clean headlines lack, the loading is attributable to the $\rho P_{i,t}$ term of (1). The flag is a proxy for $P_{i,t} \neq 0$ at the headline level; it is not derived from (1) and its validity is an empirical matter, which is why H4 is reported as a split-sample difference test rather than as an identity. The first contrast is contaminated by call-level noise, which is why the audit benchmarks the observed named-versus-anonymized dispersion against the repeat-call dispersion from the second contrast: only an excess of disagreement over the noise floor is evidence about ι. This benchmarking is, to our knowledge, new to the LLM-finance setting. The revision adds two contamination routes the initial design missed, both now measured rather than merely flagged. First, when the two conditions are fielded in different *sittings*, the contrast also absorbs session-level drift η that enters

every score produced in a given sitting. The audit's own H2 estimate was initially confounded this way —the named and anonymized rounds were twelve hours apart, and the drift alone exceeds the published shift (Section 5.2) —which is why the same-session interleaving that repairs it is part of the protocol this paper ends up recommending. Second, and discovered only when the repair experiment was run: the drift η is itself two-thirds a *batch-composition* effect —the same headline scores differently depending on which other headlines share its batch —rather than a property of the serving endpoint (Section 5.2). The score of a headline is therefore a function of the text, the firm identity, the realized prices the text describes, the call, the sitting, and the batch: the audit measures each of the last three, and none of them is a property of the text.

For the return side, write the predictive regression of interest as

$$r_{i,t+1} = \alpha + \beta\, \bar{S}_{i,t} + \gamma' X_{i,t} + u_{i,t}, \tag{4}$$

where $\bar{S}_{i,t}$ is the firm-day mean of the relevant condition's scores. Two consequences of (1) discipline the interpretation of $\hat{\beta}$.

Remark 1 (Attenuation and contamination, under explicit assumptions). Write $\bar{S}_{i,t} = \kappa\theta_{i,t} + \iota G_i + \rho P_{i,t} + \bar{\varepsilon}_{i,t}$ with $\bar{\varepsilon}$ independent of all other terms, and $r_{i,t+1} = b\,\theta_{i,t} + v_{i,t+1}$. Two restrictions make the statements below exact:

A1. G_i is uncorrelated with $\theta_{i,t}$ and with $r_{i,t+1}$ conditional on the controls $X_{i,t}$.

A2. $P_{i,t}$ forecasts $r_{i,t+1}$ only through short-horizon continuation–reversal dynamics (a coefficient b_P, say), not through θ.

A1 and A2 are *testable restrictions, not maintained hypotheses*: H2 tests A1 and H4 tests A2, and Section 5 reports the verdicts rather than assuming them. Under A1–A2, and additionally in the pure errors-in-variables case $G = P = 0$: (i) $\hat{\beta} \to b\kappa\lambda$ with $\lambda = \mathrm{Var}(\kappa\theta)/\mathrm{Var}(\bar{S})$ the reliability ratio —the classical attenuation formula, which is why H3's ICC is a first-order input into $\hat{\beta}$'s magnitude. Outside that special case: (ii) the price channel adds a contribution to the probability limit of $\hat{\beta}$ proportional to $\rho\, b_P\, \mathrm{Var}(P)/\mathrm{Var}(\bar{S})$ —exact only under mutual orthogonality of the components, which the framework of Section 3 deliberately does not impose, but directionally valid without it: an echo can generate predictive loading with no semantic content at all; and (iii) the identity term shifts the probability limit of $\hat{\beta}$ only insofar as G_i correlates with $r_{i,t+1}$ conditional on $X_{i,t}$ —exactly the object H2's predictive-difference test measures. We call this a remark rather than a proposition because its clauses are signposts with explicitly stated scope, not derivation output: the earlier draft stated them as an unproven proposition whose implicit orthogonality contradicted the framework's own "no orthogonality" stance, an inconsistency caught in audit.

The value of Remark 1 is signposting, not depth: clause (i) says repeat-call reliability is a first-order input into interpreting $\hat{\beta}$'s magnitude (H3); clause (ii) says the echo channel can generate a positive $\hat{\beta}$ with no semantic content at all, which is exactly what the H4 split tests; and clause (iii) predicts the H2 outcome we observe —no predictive shift in any of three contrasts of increasing design strength, with the level shift now measured at about three points by the interleaved design —whenever firm priors are conditionally small in their return correlation. The same-day regression $r_{i,t}$ on $\bar{S}_{i,t}$ deserves its own remark: under (1), a large same-day loading is consistent both with prices moving on news *and* with $P_{i,t}$ entering both sides. Observing that anonymization leaves the same-day loading intact (Section 5.4) is what identifies the loading as a text property.

3.3 Coverage selection

The final channel operates before any scoring happens. Let $D_{i,t} \in \{0, 1\}$ indicate that firm i has a covered firm-day at t. If

$$\Pr(D_{i,t} = 1 \mid |r_{i,t}|,\ \sigma_i) \;=\; \Phi\big(\delta_0 + \delta_1 |r_{i,t}| + \delta_2 \sigma_i\big)\,, \tag{5}$$

then the regression sample is drawn with intensity increasing in realized activity. The estimand of (4) is the conditional relationship *within covered firm-days*; extrapolation to the full panel requires $\delta_1 = \delta_2 = 0$,

which H5 tests and rejects. We emphasize that this is a statement about *what is estimated*, not an assertion that the estimate is "wrong": event-conditioned predictability is a legitimate object, but it is a different object from population predictability, and conflating the two is one of the quiet ways small-sample LLM studies overstate scope.

3.4 Inference standard

All regressions use two-way (firm and trading-day) clustered covariance following the short-panel convention; with 35 firms and 59 days, asymptotics are marginal, so H1 additionally reports a Rademacher wild-cluster bootstrap (day-level, null-imposed, $B = 1{,}999$), instrument-difference tests use day-cluster bootstrap resampling ($B = 999$), and the declared family (five hypotheses, seven primary comparisons) is reported with Benjamini–Hochberg false-discovery-rate q-values alongside raw p-values. The family itself was fixed after the first-round results were known but before any of the augmented inference above was run; this is within-study registration, not independent pre-registration, and we flag the distinction explicitly. We adopt the convention —violated in the earlier draft of this project and, we suspect, elsewhere in the literature —that the difference between two instruments is never inferred from the separate significance of their coefficients.

4 Data, Scoring Conditions, and Empirical Design

4.1 Corpus and price panel

The corpus and panel are identical to the earlier draft, so we summarize and direct the reader to the replication package for line-level provenance. Thirty-six of the largest U.S. firms across ten sectors were followed over July 1 to October 14, 2026 (74 trading days). Daily OHLCV comes from a public market-data interface (NASDAQ), yielding a balanced 2,664-firm-day price panel with no missing closes; returns are close-to-close, with same-day, overnight, and next-day open-to-close variants constructed per firm. Firm-specific news headlines were collected through a documented web-search protocol with query templates of the form "{company} stock news {month} 2026" applied uniformly across firms; the raw corpus of 542 unique URLs was cleaned by removing quote/chart pages, requiring the headline to reference the company, deduplicating by URL, and normalizing publication dates. 341 headlines are dated inside the window and average 8.4 words. The aggregation arithmetic from headlines to the analysis sample is fully explicit: the relevance filter defined below retains 335 of the 341; these occupy 276 unique firm–date cells; 36 of those cells fall on non-trading dates (weekends and holidays) and are dropped without roll-forward, leaving 240 firm-days across 36 firms and 70 of the 74 trading days. We deliberately do not roll non-trading-date news forward to the next trading day: headlines carry no release timestamps, so the publication moment relative to the nearest close is unknowable, and roll-forward would impose an artificial alignment choice. The 240 firm-days are exactly the analysis sample before the regression-level availability filters of Section 4.2.

Two properties of this corpus matter for everything that follows and are stated here rather than buried. First, the collection mechanism is a *search engine at collection time*, not a historical news wire: it oversamples retrospective coverage, market-summary articles, and prediction pieces —content that is precisely prone to describing realized prices. Section 5.4 exploits rather than ignores this property. Second, coverage is thin (9.0% of firm-days) and, as Section 5.5 shows, selected on realized activity. We treat the corpus as what it is —a search-visible slice of firm-specific news —and audit accordingly.

4.2 Four scoring conditions on the identical headline set

All LLM scoring uses the same batch protocol: the system prompt of Appendix A (reproduced verbatim), batches of 25 headlines, strict-JSON parsing with up to three retries per batch, and the same scoring model accessed through the same API with default decoding parameters (no temperature override; the API

default is not documented as deterministic, which is itself a testable property —H3). The four conditions are:

1. **Named (original).** Each headline is prefixed with its company ("[Apple] …"), preserving firm identity. Executed once on all 542 headlines in the original study; the 341 in-window scores are reused here without re-scoring.
2. **Anonymized.** Company names and tickers are replaced by "the company" in each headline (a case-insensitive alias table covering names, common short forms, and tickers; 95.3% of in-window headlines are altered), and the company prefix is removed. Everything else —prompt, batch size, parsing —is unchanged. Executed fresh on the 341 in-window headlines.
3. **Repeat calls.** The named condition is executed three additional independent times on the same 341 headlines (identical prompt, batching, and parsing), providing four draws in total (original plus three repeats) for the reliability analysis.
4. **Dictionary.** The Loughran–McDonald Master Dictionary word lists (2011 categories; 2,355 negative, 354 positive words) applied to the same headlines: tone is positive-minus-negative counts per hundred words.

All raw responses, batch logs, and the alias table are archived in the replication package; the October-16 batch log carries millisecond timestamps, while the October-15 round's log records progress without in-line timestamps and its sitting times are taken from file modification times (Appendix F). The anonymization and repeat rounds ran in a single 2026-10-16 sitting (04:24–04:33 UTC), twelve hours after the original round (2026-10-15, 16:01–16:24 UTC), under a protocol that is verbatim-identical for the named conditions; the between-session drift this gap admits is measured in Section 5.2 and Appendix F.

The interleaved replication session (2026-10-18). Once the session-confounding problem was diagnosed, the revision problem list pre-specified a confirmatory experiment —and the experiment was then run as specified. A third sitting re-fielded both conditions on the identical 341 headlines in a single session: four rounds of the named protocol and four of the anonymized protocol, interleaved at the batch level (within each round, even-numbered batches call named-then-anonymized, odd-numbered anonymized-then-named, so call order is balanced across conditions), with every call logged with timestamps and model identification (`glm-4-plus`). The same sitting also scored a composition control: the identical 341 headlines embedded in the original 542-article batch structure (v1 order and batch membership), isolating the batch-composition component of the cross-session drift from the endpoint component (Section 5.2). A fourth sitting, run later the same day, executed the mechanism discrimination pre-specified in the problem list: the 542-composition pass first (run order inverted so time-of-session works against the effect), the standard 341 pass second, and a seeded random shuffle of the 341 third (Appendix G). The session-3 design is the protocol this paper ends up recommending, executed on itself; all artifacts (twelve score files, the interleaved logs, the shuffle order, and the pre-specification) are in the replication package, and the analysis is `analysis_v3.py` / `analysis_v4.py` / `reinference.py`.

4.3 Aggregation and the analysis sample

Firm-day sentiment aggregates are means of article-level scores among headlines with relevance ≥ 30 (the LLM's own 0–100 relevance judgment, protocol-defined), matching the earlier draft exactly. The analysis sample is the 240 relevance-qualified firm-days; regressions additionally require non-missing forward returns and controls, leaving $N = 205$ firm-days (35 firms, 59 days) for the core specifications. Controls are the lagged firm return, lagged 21-day annualized volatility, and the equal-weighted market return of the 36-firm panel. Standard errors are clustered by firm and trading day throughout; augmented inference (wild bootstrap, difference bootstraps, BH family) is as specified in Section 3.

4.4 Pre-specified test family

The audit evaluates the five hypotheses H1–H5 of Section 2 over a declared family of seven primary comparisons: H1 (named next-day coefficient), H2 (score-level shift; predictive-coefficient difference named−anonymized; predictive-coefficient difference named−dictionary), H3 (ICC), H4 (price-language split difference), and H5 (coverage-selection test). Raw p-values and Benjamini–Hochberg q-values are reported side by side in Table 4. Auxiliary statistics —horizon paths, portfolio sorts, subsample splits, and interaction terms —are reported as descriptive or secondary evidence and are not part of the declared family; where the earlier draft presented such statistics as findings, we note the re-interpretation explicitly.

4.5 Descriptive statistics

Table 1 reports article-level and firm-day-level summary statistics for all conditions; every cell is computed by `descriptives.py` from the archived scores and panel, so the table and the text below cannot drift apart. Three contrasts deserve emphasis. First, the named and anonymized distributions have similar dispersion (SD 39.0 vs. 36.4 at the article level) but different centers (means +16.9 vs. +12.1): anonymization moves a level, not a shape. Second, the dictionary remains a sparse instrument —non-zero for 13.5% of headlines versus 70.1% for the LLM —and its firm-day aggregate is non-zero for only 16.3% of the analysis sample, an attenuation issue for any dictionary comparison in short texts (Loughran and McDonald, 2011). Third, the three repeat draws are individually similar in marginal distribution (means spanning 10.5 to 12.0 points) yet stochastic call-by-call: only 29.9% of headlines receive the identical score across all four draws, and the pairwise exact-match rate averages 42.4% across cross-session pairs versus 56.9% within the October 16 sitting —the signature of a reliable-in-rank but noisy instrument, whose disagreement is amplified across sittings (Sections 5.3 and 5.2).

Table 1: Descriptive statistics across scoring conditions

	Mean	Std. dev.	Min	Max	N	Nonzero
Panel A: Article level (341 in-window headlines)						
LLM sentiment, named (−100 to +100)	16.88	38.97	-80	90	341	70.1%
LLM sentiment, anonymized	12.13	36.45	-80	90	341	60.1%
LLM sentiment, repeat draws 1–3 (ranges)	10.45–11.98	34.99–36.85	—	—	341	—
LM dictionary tone (per 100 words)	0.25	4.68	−20.0	20.0	341	13.5%
Price-language headline flag	0.152	—	0	1	341	—
Relevance (0–100)	72.0	18.0	0	100	341	—
Panel B: Firm-day level (240 relevance-qualified firm-days)						
Named sentiment ($[-1, 1]$ scale)	0.14	0.36	−0.80	0.85	240	70.8%
Anonymized sentiment	0.10	0.34	−0.80	0.85	240	62.5%
Dictionary tone ($\times 10^{-2}$)	0.19	4.39	−20.0	20.0	240	16.2%
Headlines per firm-day	1.25	0.72	1	7	240	—
Next-day return (bps)	22.0	320.5	-915	2258	234	—
Lagged 21-day volatility (annualized)	0.380	0.200	0.13	1.12	240	—

Panel A is computed over the 341 in-window headlines; Panel B over the 240 relevance-qualified trading-day firm-days of the analysis sample (next-day returns are non-missing for 234 of 240 firm-days). Repeat draws report ranges across the three independent calls. Nonzero is the share of observations with score $\neq 0$. Every cell is generated by `descriptives.py` from the archived scores and panel; the previous hand-transcribed version mixed an all-542 computation into the $N = 341$ rows and has been withdrawn.

Figure 1 summarizes the full audit design: corpus, four conditions, the decomposition of equation (1), and the mapping from channels to hypotheses.

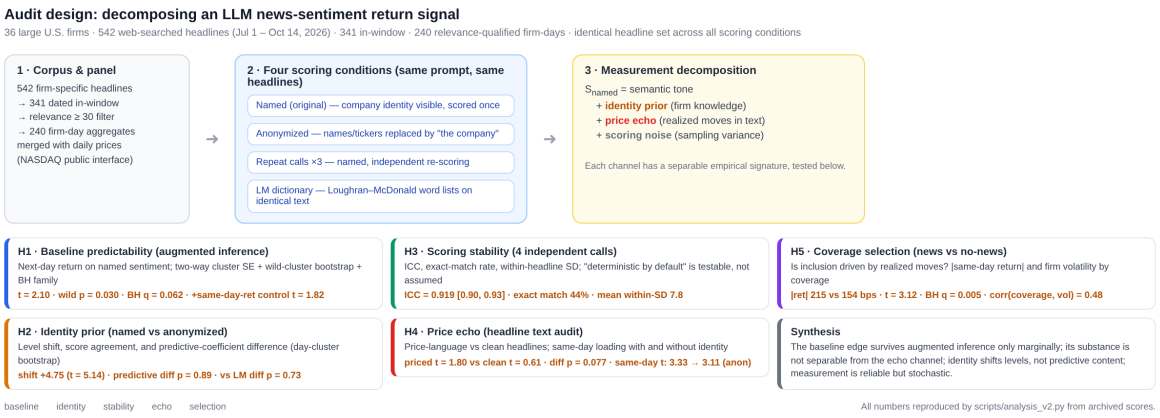


Figure 1: Audit design. One frozen corpus is scored under four conditions (named, anonymized, three repeat calls, dictionary), each isolating one term of the measurement decomposition $S = \kappa\theta + \iota G + \rho P + \varepsilon$; the five pre-specified hypotheses test the channels jointly under a common inference standard. All numbers are reproduced by `analysis_v2.py` from archived scores.

5 Audit Results

5.1 H1: baseline predictability under augmented inference

Table 2 column (1) reports the pre-specified baseline: next-day close-to-close returns on named firm-day sentiment with controls, two-way clustered. The coefficient is 0.0149 ($t = 2.10$, $p = 0.035$), i.e., a one-standard-deviation sentiment increase (0.36 on the $[-1, 1]$ scale) is associated with a 54 bps next-day return. Three augmentations discipline this number. First, the Rademacher wild-cluster bootstrap (day-level, null-imposed, $B = 1{,}999$) gives $p = 0.030$: the result is not an artifact of one or two trading days. Second, the Benjamini–Hochberg correction over the declared seven-comparison family raises the implied q-value to 0.062: the headline baseline would not survive a 5% FDR threshold. Third, adding the same-day return as a control —motivated by the echo analysis below —reduces the coefficient to 0.0141 with $t = 1.82$ ($p = 0.069$); adding the overnight component as well gives $t = 1.71$. The honest reading of H1 is *marginal evidence, sensitive to specification, in a 205-observation sample*. The design's resolution is similarly bounded: at 80% power, the minimum detectable next-day coefficient is 0.020 —one-third larger than the estimate itself —and the power of the test against the observed effect is 0.56 (Appendix E). We flag that the earlier draft of this project characterized the same statistic as a finding; the augmented inference supports only the weaker claim.

For comparability with the literature, the same-day coefficient is 0.0283 ($t = 3.33$, $N = 205$): sentiment co-moves with the return realized *on* the news day. In the earlier draft this was interpreted through a price-formation model (the "$\lambda\gamma$" term). Section 5.4 shows why that interpretation was premature: the loading is a property of the text, not evidence about how prices form.

One further augmentation was added once the interleaved replication session of Section 5.2 became available: *draw stability*. Re-estimating the identical H1 specification with every archived scoring draw of the same headlines —the October-16 repeat mean, four October-18 named rounds and their average, the October-18 five-hundred-article composition pass, and both anonymized aggregates —yields coefficients in $[0.0116, 0.0161]$ and t-statistics in $[1.47, 2.12]$ (the October-16 single repeats individually run $t = 2.12, 1.76, 1.68$; Section 6). The point estimate is stable to well within its own standard error, but the *nominal significance of H1 is draw-dependent*: the published draw sits at the favorable end of the range, and a pipeline that scored once on a different day would have reported p between 0.05 and 0.14. We accordingly read H1 as *borderline evidence with a stable point estimate*, which is exactly the resolution the power analysis of Appendix E predicts for a true effect of this size.

Table 2: Core audit regressions (dependent variable: next-day return unless noted)

	(1) Named	(2) Named	(3) Anon.	(4) Dict.	(5) Price-lang.	(6) Clean
Sentiment (named)	0.0149*	0.0141			0.0470	0.0033
	(2.10)	(1.82)			(1.80)	(0.61)
Sentiment (anonymized)			0.0144			
			(1.78)			
Dictionary tone				0.0268		
				(1.00)		
Same-day return		0.0306			0.0961	−0.0211
		(0.38)			(1.13)	(−0.29)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	205	205	205	205	38	167
Wild-cluster bootstrap *p*	0.030	—	—	—	—	—
BH family *q*	0.062	—	—	—	—	—

All specifications include lagged return, lagged 21-day volatility, and the equal-weighted market return; *t*-statistics in parentheses, clustered by firm and trading day. Columns (5)–(6) split the sample by whether any headline for the firm-day contains realized-price language (15.2% of headlines); both include the same-day return as a control. Wild-cluster bootstrap: Rademacher weights at the day level, null imposed, $B = 1{,}999$. BH *q*: Benjamini–Hochberg FDR over the declared family of Table 4. *10%, **5%, ***1%.

Figure 2 presents all core coefficients with 95% confidence intervals: the audit's message in one picture is that every instrument-condition coefficient is positive, every interval is wide, and the differences among them are the object of the next subsection.

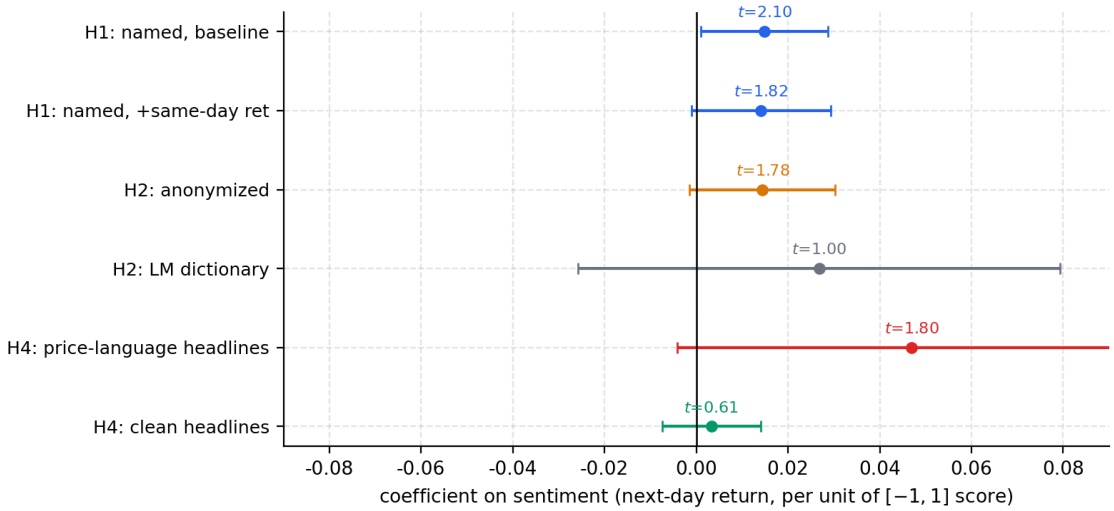


Figure 2: Core coefficients with 95% confidence intervals (two-way clustered). The named, anonymized, and dictionary conditions are statistically indistinguishable from one another (Table 3); the predictive loading is concentrated in price-language headlines.

5.2 H2: identity shifts levels, not predictive content

Measurement layer. On the identical 341 headlines, named scores exceed anonymized scores by 4.75 points on average (paired $t = 5.14$, $p < 10^{-6}$). The two distributions correlate at 0.900 (Pearson; Spearman 0.888); 43.4% of headlines receive the identical score in both conditions, and the mean absolute difference is 11.2 points. This is what the audit originally reported as an identity prior, and it is what the revision re-opens. The two conditions were not fielded in the same sitting: the original named round ran on 2026-10-15 (16:01–16:24 UTC), while the anonymized round and all three named repeats ran

in a single 2026-10-16 sitting (04:24–04:33 UTC), roughly twelve hours later, under a protocol that is verbatim-identical for the named conditions (same system prompt, same user-message format, same batch size). The named protocol itself drifted +5.79 points in level between the two sittings (paired $t = 7.03$, $p < 10^{-11}$): the same prompts, the same headlines, twelve hours apart, score six points more bullish in the first sitting. Benchmarked within session, the identity level effect is −1.04 points in the named-minus-anonymized direction —anonymized scores sit *above* their same-session named twins, with the mean test not reaching significance ($t = 1.60$, $p = 0.11$) while the rank test does (Wilcoxon $p = 0.04$; against the three repeats individually, +1.30, +1.67, and +0.15 in the anonymized direction, one of three below 0.05). The disclosure of a de-identification failure sharpens this further: the executed alias table contains no "Visa" entry, so for the ten ticker-V headlines the anonymization was a no-op —the "anonymized" text still names the company (Appendix B) —and excluding those ten headlines moves the same-session shift to −1.29 ($t = -2.05$, $p = 0.04$; Wilcoxon $p = 0.03$). The honest summary is a range, not a verdict: a same-session anonymization displacement of roughly one point, negative in the named-minus-anonymized direction, whose significance depends on the test and on the V exclusion —an order of magnitude smaller than the session drift either way. The published shift decomposes exactly: +4.75 = +5.79 (session drift) −1.04 (same-session identity). Figure 4 displays the decomposition. We accordingly retract the interpretation of the +4.75 shift as a firm-specific prior: on this corpus, the scorer's apparent optimism about known firms is predominantly the scoring process moving between sittings, and the earlier framing of the shift as the out-of-sample analog of Glasserman and Lin (2023)'s distraction effect was an artifact of session scheduling. What survives is a protocol lesson with teeth: condition contrasts fielded in separate sittings absorb session-level drift, and only same-session interleaving identifies the contrast the design intends. The dispersion benchmarks agree with the level evidence: within the October-16 sitting, anonymized-versus-named displacement (7.9 points of mean absolute difference) exceeds repeat-versus-repeat displacement (6.5 points) by roughly the same one-point margin as the level shift, while the cross-session displacement (11.3 points) towers over both.

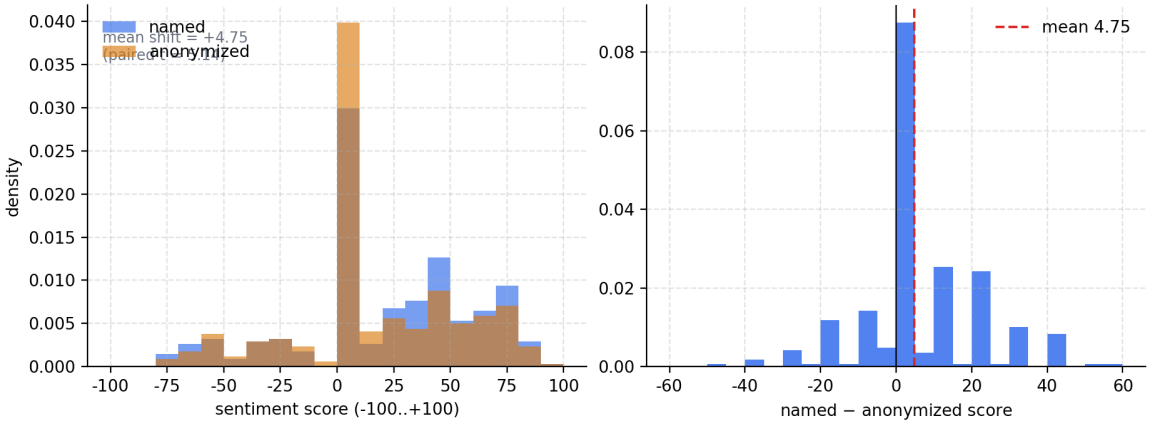


Figure 3: Named versus anonymized score distributions on the identical 341 headlines. Left: overlapping histograms (the named distribution sits to the right). Right: histogram of paired differences, with the mean shift of +4.75 marked. Both panels contrast the October-15 named round with the October-16 anonymized round and therefore mix identity with session drift; see Figure 4.

The confirmatory experiment. The same-session contrast above is itself limited in two ways the revision problem list made explicit before the experiment was run: the October-16 sitting was *not* interleaved (all anonymized batches were scored before the named repeats, so within-sitting drift is absorbed into the contrast), and each condition was measured with a single draw per headline. A third sitting (2026-10-18, endpoint reported by the API metadata as `glm-4-plus`) therefore re-fielded both conditions under the design this paper ends up recommending: four rounds, within each round the fourteen batches of 25 alternate condition order (even batches named-then-anonymized, odd batches anonymized-then-named), protocol verbatim-identical. The design was pre-specified in the revision problem list before data collection and is reported here as a confirmatory contrast outside the declared family, well-powered under both headline-

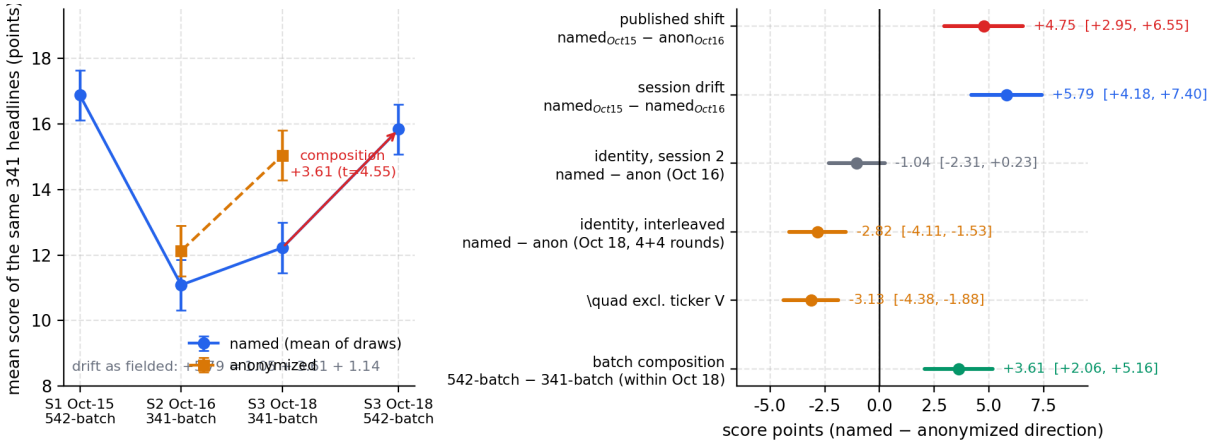


Figure 4: The level anatomy of the score across three sittings and two batch compositions. Left: mean score of the identical 341 headlines in each cell —the drift as originally fielded (+5.79, session 1 at the 542-batch composition to session 2 at the 341-batch composition) decomposes exactly into an endpoint leg (+1.05), the batch-composition leg (+3.61, measured within session 3), and the second endpoint leg (+1.14). Right: all identity, drift, and composition contrasts with 95% CIs —the interleaved session establishes the identity effect at −2.82 where the non-interleaved session-2 contrast could only suggest it.

level and call-clustered inference. The result settles what the session-2 contrast could only suggest: the interleaved identity effect is −2.82 points in the named-minus-anonymized direction, with every one of the four round-level estimates negative (−3.99, −2.05, −2.74, −2.51; round-level $t = 6.8$, $p = 0.007$). One inference discipline discovered in producing this draft applies here and below: because each API call returns all 25 scores of a batch in a single response, the scores of one call share a response-level *mood* (Section 5.3 quantifies it at 13% of residual variance), so headline-level paired tests overstate precision; under the most conservative clustering (14 batch-calls) the identity effect retains $t = 2.7$ ($p = 0.017$; 95% CI [−4.85, −0.79]), and the V-excluded estimate −3.13 retains $t = 4.8$ ($p = 0.0003$). *Anonymized scores sit systematically above their named twins by about three points*: the identity prior is real, negative, and roughly triple the magnitude the underpowered session-2 contrast suggested —the opposite sign from the published +4.75, which session drift had masked entirely. The order alternation makes call-order contamination testable: the identity effect is 3.34 points (in the anonymized-minus-named direction) on batches where the named call came first versus 2.27 where it came second —similar magnitudes, no order artifact. The paired exact-match rate between interleaved condition calls is 48.5%, bracketed by the within-condition repeat rates, exactly as a level displacement with stochastic noise predicts.

What the drift is made of. The interleaved session also closes the one question the two-session evidence could not answer: whether the +5.79 cross-session drift reflects the serving endpoint or the change in batch composition between sittings (session 1 scored the headlines embedded in the full 542-article stream; session 2 scored them alone). The October-18 sitting includes a control pass for exactly this: the *identical* 341 headlines scored under the original 542-article batch composition, in the same session, same endpoint, same order. The composition effect is +3.61 points (headline-level paired $t = 4.55$; batch-call-clustered $t = 2.74$, $p = 0.012$), and a fourth sitting re-ran the control with the run order *inverted* —composition pass first, 341-pass last, so that time-of-session works against the effect —and reproduces it at +3.50 (clustered $t = 2.17$); the pooled two-session estimate is +3.56 ($z = 3.4$). Together with the two fixed-composition session legs, this gives an additive decomposition of the published drift: $+5.79 \approx +1.05$ (session 1 minus session 3, both at the 542 composition) + 3.61 (composition, within session 3) + 1.14 (session 3 minus session 2, both at the 341 composition), with the endpoint legs non-monotonic and the cross-session drift at fixed composition measuring +0.57 ($t = 0.3$) between sessions 3 and 4. Batch composition, not endpoint change, carries roughly 62% of the apparent "session drift" —which means the score of a headline depends measurably on *which other headlines are scored alongside it*, a measurement-validity channel we believe has not previously been quantified. The fourth sitting also randomized the batch assignment of the 341 (seeded shuffle): re-batching the *same* corpus moves levels

by +4.72 points (clustered $t = 2.02$, $p = 0.065$ —suggestive), and the mechanism work of Appendix G shows why this is subtle: linear anchoring on batch-mates' fixed content is estimated at $\theta = +0.36$ ($t = 4.9$) but *provably* cannot shift the corpus mean (any re-batching preserves the average of batch-mate means); the shifts instead require the non-linearities we detect —anchoring is asymmetric (positive batch-mates assimilate at +0.69, $t = 2.3$; negative ones do nothing) —together with the response-level mood. For pipelines the operational consequence is sharper than the session rule: hold the batch composition fixed across any comparison —across conditions, sessions, or model versions —because re-batching alone moves levels by more than the effect most studies are trying to measure.

Prediction layer. Column (3) of Table 2 re-estimates the baseline with the anonymized aggregate: the coefficient is 0.0144 ($t = 1.78$) against the named 0.0149 ($t = 2.10$). The formal day-cluster bootstrap difference test (Table 3) gives an observed difference of 0.0004 with 95% CI $[-0.0044, 0.0045]$ and $p = 0.887$: the two conditions are statistically indistinguishable as predictors. Because that comparison pairs an October-15 named score with an October-16 anonymized score, we re-run it entirely inside the October-16 sitting —the anonymized regression is the same one either way; what changes is the named side of the contrast —so the mean of the three same-session named repeats (0.0147, $t = 1.87$) is compared against the same anonymized aggregate (0.0144), difference 0.0002 (day-cluster bootstrap $p = 0.905$). The interleaved session supplies the definitive version of the test, free of both the session and the batch-composition confounds at once: the October-18 four-round named average gives 0.0146 ($t = 1.67$) against the same-session anonymized average 0.0133 ($t = 1.88$), difference +0.0013 with 95% CI $[-0.0034, 0.0074]$ and $p = 0.622$ ($B = 9{,}999$). The prediction-layer conclusion —identity does not change predictive content —is now confirmed under three contrasts of increasing design strength: cross-session, same-session, and same-session interleaved with replication. The named-versus-dictionary difference is likewise insignificant (-0.0119, CI $[-0.0737, 0.0576]$, $p = 0.731$). In a horse race on the same observations (not shown to avoid triple-counting), both coefficients remain individually insignificant at 5%, which is exactly what low power plus attenuation predicts (Remark 1, clause (iii)).

The two-layer result is the paper's cleanest inference about what the model brings to the text —now corrected twice by its own revisions and confirmed by the interleaved design: identity moves the level by about three points (a mild pessimism prior: named scores sit *below* their anonymized twins), moves levels by another three-plus points through batch composition, and does not detectably move the predictive signal in any comparison, including the definitive interleaved one. A prior that inflated all of a firm's headlines equally would be differenced out of any predictive regression that conditions on the firm; a prior that differentially tracked which headlines *matter* would have shown up as a predictive difference, and does not. The level evidence that initially pointed the other way measured the session and the batch, not the prior —and the sign it hid was the opposite of the one reported.

5.3 H3: the instrument is reliable but stochastic

Four independent calls of the identical prompt on the identical headlines yield an intra-class correlation $\mathrm{ICC}(2, 4) = 0.919$ with a bootstrap 95% CI of $[0.90, 0.93]$; mean pairwise Pearson correlation across draws is 0.928; 29.9% of headlines receive the identical score in all four calls (44.0% for the original-versus-draw-1 pair alone), and the mean within-headline standard deviation is 7.8 points (Figure 5). Those four calls span two sittings; the three October-16 draws alone give $\mathrm{ICC}(2, 3) = 0.949$, so pooling across the twelve-hour gap costs three reliability points. The interleaved session settles the natural follow-up question —is the October-16 reliability itself a fluke of that sitting? Four October-18 named rounds give $\mathrm{ICC}(2, 4) = 0.949$ with a within-headline standard deviation of 7.2 points (6.6 for the three October-16 rounds), and the interleaved anonymized rounds match (0.949): reliability is a stable property of the instrument, not of a particular sitting, even while the *level* moves by points between the same sittings. The eight passes of the interleaved design also let the noise itself be decomposed, and the decomposition revises the intuition a single-repeat design encourages: 13% of the residual variance is a *response-level mood* —a random level shared by all 25 headlines of one API response (SD $\approx$ 2.6 points) —and only the remaining 87% is idiosyncratic per headline per response (SD $\approx$ 6.7). Scoring is not 341 independent draws per pass but 14 correlated clusters, which is why the level contrasts of Section 5.2 are reported

Table 3: Formal difference tests (day-cluster bootstrap, $B = 999$)

Contrast	Observed difference	95% CI	p-value
$\beta_{\text{named}} - \beta_{\text{anonymized}}$ (cross-session)	0.0004	$[-0.0044,\ 0.0045]$	0.887
$\beta_{\text{named}} - \beta_{\text{anonymized}}$ (same session)	0.0002	$[-0.0034,\ 0.0037]$	0.905
$\beta_{\text{named}} - \beta_{\text{anonymized}}$ (interleaved)	0.0013	$[-0.0034,\ 0.0074]$	0.622
$\beta_{\text{named}} - \beta_{\text{dictionary}}$	−0.0119	$[-0.0737,\ 0.0576]$	0.731
$\beta_{\text{price-lang.}} - \beta_{\text{clean}}$	0.0436	$[-0.0059,\ 0.1105]$	0.100
Level shift, cross-session (points)	+4.75	—	<0.0001
Session drift (points)	+5.79	—	<0.0001
Identity, same session (points)	−1.04	$[-2.32,\ +0.23]$	0.111
Identity, same session, excl. V (points)	−1.29	$[-2.53,\ -0.05]$	0.041
Identity, interleaved sess. 3 (points)	−2.82	$[-4.85,\ -0.79]$	0.017
excl. ticker V	−3.13	$[-4.41,\ -1.86]$	0.0003
Composition effect, sess. 3 (points)	+3.61	$[+1.03,\ +6.20]$	0.012
Composition effect, sess. 4, order inverted	+3.50	—	0.042
Re-batch (random), sess. 4	+4.72	—	0.065

Differences of next-day predictive coefficients from Table 2 specifications; bootstrap resamples trading-day clusters ($B = 999$; the price-language row uses $B = 9{,}999$ under the same-day-control specification actually estimated, and is seed-stable: three independent seeds give p between 0.093 and 0.100; the interleaved row uses $B = 9{,}999$). Level rows are paired t-tests at the article level ($n = 341$, points on the $[-100, 100]$ scale): cross-session rows compare the 2026-10-15 named round with the 2026-10-16 round; the same-session identity rows compare the anonymized condition against the mean of the three October-16 named repeats. The last five rows are the revision-stage confirmatory contrasts of the interleaved sessions (2026-10-18 and a mechanism sitting; Section 5.2 and Appendix G), pre-specified outside the declared family and reported under batch-call-clustered inference because the 25 scores of one API response share a response-level mood (13% of residual variance; Section 5.3) —headline-level tests of the same contrasts give $t = 4.28$ (identity) and $t = 4.55$ (composition), and the round-level identity meta-test gives $t = 6.8$ ($p = 0.007$). The V exclusion drops the ten headlines whose anonymization was a no-op (Appendix B); the composition rows compare the identical 341 headlines scored inside the original 542-article batches versus the 341-only batches of the same session (session 4 with the run order inverted); the re-batch row compares seeded-random batches to the standard order. "One significant, the other not" never appears in this table by construction: all cross-instrument statements are direct difference tests. Computed by `analysis_v2.py` (family rows), `revision_power.py` and `v_exclusion.py` (session rows), and `analysis_v3.py` / `analysis_v4.py` / `reinference.py` (session-3/4 rows).

under call-clustered inference and why a pipeline that averages multi-headline batches inherits a noise floor its headline-level statistics hide. The pooled number remains the right one to report for a production pipeline —single-sitting ICCs overstate what re-scoring tomorrow will deliver —and the same pooling is why the published ICC already prices in the session drift documented in Section 5.2. Reliability of 0.92 is high by psychometric standards for a single-item instrument, and the mean-level agreement across draws (within one point) rules out gross instability. But it is emphatically *not* determinism: a research pipeline that scores each headline once and treats the result as "the" score embeds a measurement-noise floor of ±8 points per headline into every downstream construct. Under Remark 1(i), the observed coefficient is attenuated by roughly the reliability ratio; with firm-day averaging over 1.25 headlines the attenuation is modest, which is the arithmetic reason the named and anonymized coefficients survive as similar point estimates. The practical implication is procedural: single-call scoring should report an ICC or repeat-call dispersion alongside any predictability statistic, and pipelines should score at least twice and average when headlines are cheap relative to inference —as they are here.

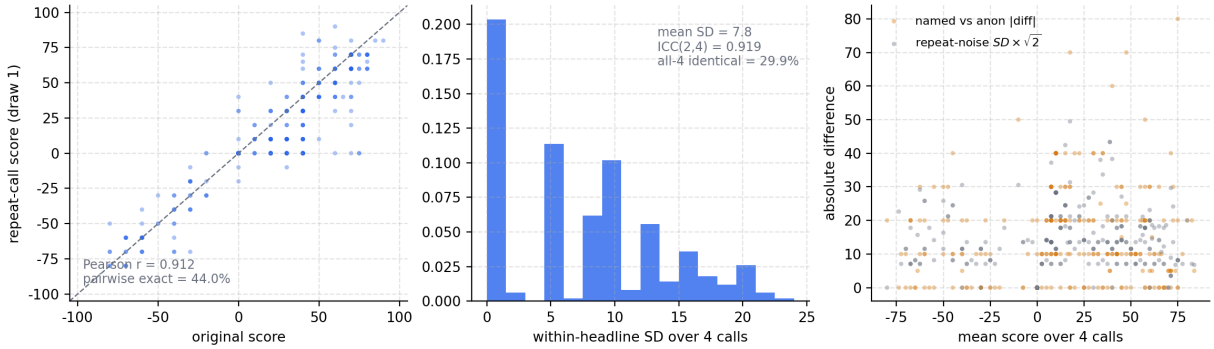


Figure 5: Repeat-call reliability. Left: original versus draw-1 scores ($r = 0.912$; 44.0% of the pair identical). Middle: within-headline standard deviation over four calls (mean 7.8 points); 29.9% of headlines receive one identical score in all four calls. Right: Bland–Altman comparison of the named-versus-anonymized absolute difference against the repeat-noise floor ($\sqrt{2}\times$ within-SD): anonymization displaces scores by more than re-calling the model does.

5.4 H4: the predictive content sits in the price echo

A text audit flags headlines containing realized-price language —movement verbs and magnitude constructions ("rose," "surged," "dropped by 5% after earnings," "up 20% in 2026") —covering 52 of 341 in-window headlines (15.2%). Table 2 columns (5)–(6) split the firm-day sample by the flag (all specifications include the same-day return as a control): the coefficient is 0.047 ($t = 1.80$, $N = 38$) for firm-days with price-language headlines and 0.003 ($t = 0.61$, $N = 167$) without. The bootstrap difference is 0.044 with 95% CI $[-0.006, 0.110]$, $p = 0.100$ ($B = 9{,}999$, seed-stable; Table 3): directional evidence that does not clear the 10% threshold, underpowered in the tails, that the predictive loading concentrates where the text describes realized moves. On the magnitudes: essentially the entire point estimate of the baseline coefficient comes from the 19% of the sample flagged as echoes; the clean subsample's coefficient is one-fourteenth of the baseline and indistinguishable from zero.

Because the split's validity rests on a regular-expression classifier, the revision audited the classifier itself: all 52 flagged headlines were re-classified one by one against a written rulebook that distinguishes realized price moves from fundamental-magnitude language ("revenue rises 12%" —the rise is the revenue's, not the stock's), directional forecasts ("could sink," "may drop to $300"), and pure regex false positives ("execution short fall"; "Walmart just dropped October deals" —retail promotions, not a price move). The classification is LLM-assisted and disclosed as such —it is an internal audit of the flag, not independent human annotation (Appendix H). The verdict: 43 of 52 (82.7%) are genuine realized-price headlines; three are fundamental-magnitude, three are forecasts, and three are false positives. Re-running the split with only the strict realized-price firm-days leaves the point estimate essentially unchanged (0.0475 versus 0.0470; $t = 1.61$, $N = 33$) —the flag was noisy at the margin but not wrong where it mattered. The

classification also sharpens the echo signature itself: strict-price firm-days average 361 bps of absolute same-day return against 178 for clean days, while the nine misclassified headlines' firm-days average 120 bps —below even the clean baseline, exactly what non-echo headlines should look like. Dominance checks confirm the split is not carried by an outlier firm or day: the largest single firm contributes 9% of the strict-price firm-days, and leave-one-out and leave-one-company-out jackknifes of the priced-minus-clean difference keep every replicate positive (sign-flip rate 0.0; Appendix H). The honest reading of H4 is unchanged by the audit —directional, underpowered, text-located —but it is now directional evidence whose classifier has itself been audited.

Two corroborating facts identify the channel as textual rather than mnemonic. First, the same-day loading of returns on sentiment is $t = 3.33$ named and $t = 3.11$ anonymized: if the same-day co-movement were driven by the scorer's memory of price outcomes, removing identity should have collapsed it; it barely moves. Second, the earlier draft's horizon path —positive at $t+1$, negative at $t+2$ ($t = -2.04$) —is exactly what short-horizon continuation-and-reversal around high-absolute-return events produces mechanically (Chan, 2003), on a sample that (per H5) over-weights exactly those events. The reversal, which the earlier draft labeled its most novel finding and read as evidence of attention-driven overshooting, is better read as the echo channel propagating through return dynamics; we return to this in Section 6.

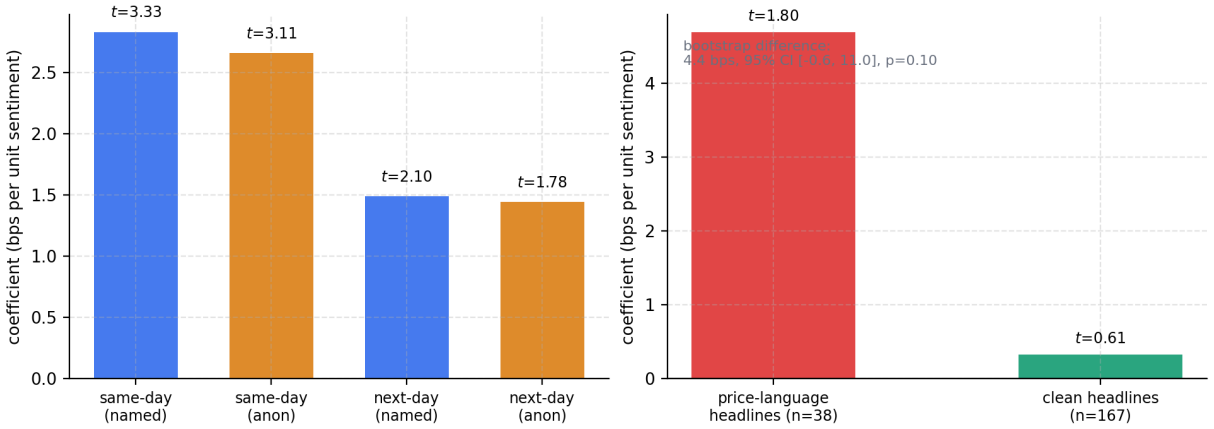


Figure 6: The echo channel. Left: same-day and next-day coefficients, named versus anonymized —anonymization does not remove the same-day loading (3.33 vs. 3.11), locating the contamination in the text. Right: next-day coefficients for price-language versus clean headlines, with the bootstrap difference.

5.5 H5: coverage is selected on realized activity

Covered firm-days exhibit mean absolute same-day returns of 215 bps against 154 bps for uncovered firm-days, and firm-level news counts correlate 0.48 with firm return volatility (Figure 7). The v2 draft tested the coverage gap with a pooled Welch statistic ($t = 3.12$, $p = 0.002$); that test ignores the double indexing of the 2,664-cell panel —the same firm repeats across days, and all firms share each trading day's market shock —so the v4 inference replaces it with a firm-and-day fixed-effects regression of absolute returns on the coverage indicator, with two-way clustered standard errors: $t = 2.76$ ($p = 0.006$; BH $q = 0.014$). Two robustness specifications agree: firm fixed effects with day-clustered errors give $t = 2.49$ ($p = 0.013$), and a day-cluster bootstrap of the gap gives $p \leq 0.002$ with 95% CI $[25, 103]$ bps. The selection equation (5) is therefore rejected with $\delta_1 > 0$ under panel-appropriate inference: the audit sample over-weights high-activity events. The earlier draft asserted that "coverage is driven by query design rather than by return outcomes; the consequence is statistical power, not bias." That assertion is wrong, and we retract it: search-driven collection selects on exactly the outcomes that make short-horizon continuation and reversal strongest for non-semantic reasons. The correction does not invalidate within-covered-sample inference, but it bounds external validity —the estimand is the conditional (event-sampled) relationship, not the population relationship —and it interacts with H4: high-activity events are both more likely to be covered and more likely to be covered *retrospectively*.

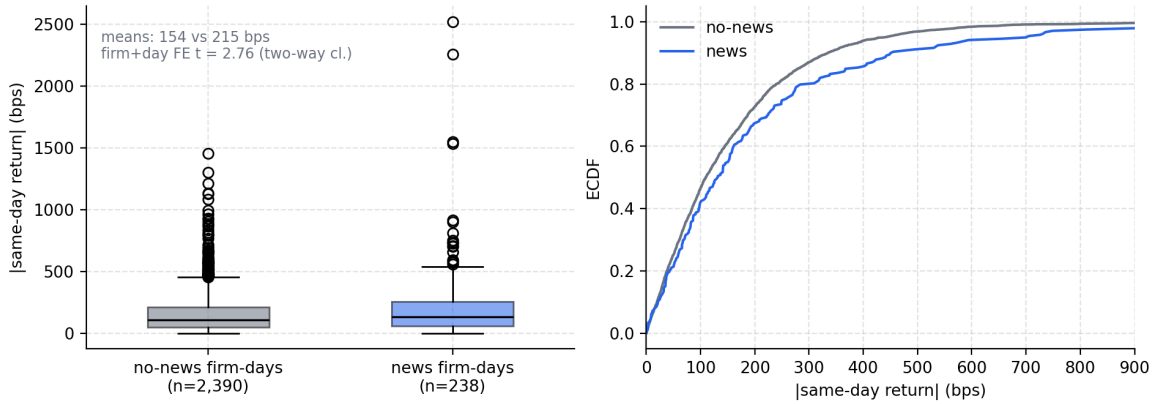


Figure 7: Coverage selection. Left: distribution of absolute same-day returns for covered versus uncovered firm-days. Right: empirical CDFs. Covered firm-days are drawn disproportionately from high-absolute-return states.

5.6 Multiple-testing summary

Table 4 reports the declared family with raw and Benjamini–Hochberg q-values. Three comparisons survive at any conventional threshold —the named-versus-anonymized level shift ($q < 0.0001$), the coverage-selection test ($q = 0.014$), and the reliability ICC ($q < 0.0001$, one-sided). The revision re-interprets the first of these: the level-shift test rejected because its contrast mixes an identity effect with a twelve-hour session gap, and the session component alone (+5.79, $t = 7.03$) would clear any threshold; it survives as evidence of *scoring instability across sittings*, not of an identity prior (Section 5.2). The two channels that would constitute "an LLM sentiment return signal" —the baseline coefficient and its echo decomposition —sit at $q = 0.062$ and $q = 0.140$ respectively. The pattern is itself informative: the audit finds the contamination and instability channels with confidence and the predictive content only marginally. The revision-stage confirmatory contrasts of Section 5.2 —the interleaved identity effect (round-level $p = 0.007$; call-clustered $p = 0.017$), its V-excluded variant ($p = 0.0003$), and the batch-composition effect (pooled $z = 3.4$, two sessions with inverted run order) —sit outside the declared family by construction (they post-date it), but they are pre-specified in the revision problem list and point the same way: the channels the audit measures with confidence are the ones the score carries for reasons other than the text's semantics.

Table 4: Declared test family: raw p-values and Benjamini–Hochberg q-values

Comparison	raw p	BH q
H2: identity level shift (named − anonymized)	<0.0001	<0.0001
H3: ICC > 0.5 (one-sided bootstrap)	<0.0001	<0.0001
H5: coverage selection (firm+day FE, two-way cl.)	0.0059	0.0137
H1: baseline next-day coefficient (named)	0.0354	0.0620
H4: price-language vs. clean difference	0.1002	0.1402
H2: predictive difference, named vs. anonymized	0.8869	0.8869
H2: predictive difference, named vs. dictionary	0.7307	0.8525

Family declared in Section 4 prior to augmented inference. The three rejections are measurement-layer channels; the two return-layer channels are marginal. q-values from the Benjamini–Hochberg step-up procedure at $m = 7$. The H2 level-shift row compares conditions fielded twelve hours apart; Section 5.2 attributes most of it to between-session drift, and the same-session contrast ($p = 0.11$, or $p = 0.04$ excluding the ticker-V no-op anonymization) is a revision-stage addition outside the declared family. The H5 row is the v4 panel specification (firm and day fixed effects, two-way clustered) that replaces the v2 pooled Welch statistic, which ignored the panel's double indexing.

6 Robustness and Threats to Validity

6.1 Specification robustness of the baseline

Five checks, all computed fresh on the archived data. (i) Raising the relevance threshold from 30 to 50 leaves the baseline essentially unchanged ($t = 2.12$, $N = 187$). (ii) Using the unweighted mean over all headlines with no relevance filter gives $t = 2.07$ ($N = 209$): the filter is not doing the work. (iii) Dropping dense event days (five or more covered firms) leaves the point estimate intact but the statistic at $t = 1.61$ ($N = 117$): about a quarter of the baseline's strength comes from clustering on event days, consistent with H5. (iv) The anonymized condition with the same-day control runs at $t = 1.59$, tracking its named twin. (v) A within-day permutation placebo (1,000 draws, sentiment permuted inside each trading day) yields a two-sided permutation $p = 0.007$: whatever the baseline coefficient is, it is not an artifact of arbitrary re-labeling. The consistent picture across (i)–(v) is a real but fragile association —robust to measurement choices, weakened by event-day down-weighting, and significant at conventional levels only under the original sample construction.

6.2 Session robustness of the predictive coefficient

The baseline coefficient rests on October-15 scores; the three October-16 named repeats re-score the identical headlines in a different sitting, and the October-18 interleaved session re-scores them twice more per condition in a third. Across the ten archived draws of the named protocol (original, three October-16 repeats, four October-18 rounds, the October-18 542-composition pass, and session means), the next-day coefficient lives in $[0.0116, 0.0161]$ and the t-statistic in $[1.47, 2.12]$: the point estimate is stable across sessions to well within its own standard error, and every repeat falls inside the range the specification checks above already span —but the *significance* of the coefficient moves with the draw, from $p = 0.035$ (the published round) to $p = 0.14$ (the weakest October-18 round). The same-day loading is large under every draw (t between 2.96 and 3.80 for the session-1/2 draws). Session drift is therefore a *level* phenomenon —it moves the mean score by about six points (Section 5.2) —not a *slope* phenomenon: predictive loadings survive re-scoring, which is the robustness fact every other result in this paper leans on. The same-session instrument difference is itself stable across all three designs: the anonymized-versus-named predictive difference is 0.0002 ($p = 0.905$) inside the October-16 sitting, 0.0013 ($p = 0.622$) inside the interleaved October-18 session, tracking its cross-session twin ($p = 0.887$).

6.3 The horizon path, re-interpreted

The earlier draft emphasized a sign pattern —positive at $t + 1$ ($t = 2.10$), negative at $t + 2$ ($t = -2.04$, $p = 0.042$), fading by $t + 3$ ($t = -1.52$) —and read it as evidence of attention-driven overshooting and correction. The audit re-interprets this pattern rather than discarding it. A subsample selected on high absolute returns (H5) and scored with text that partially echoes those returns (H4) mechanically produces short-horizon continuation followed by reversal: this is the classical news-event return dynamics documented by Chan (2003) for reasons (liquidity provision, bid–ask bounce, order-flow imbalance) that require no sentiment semantics at all. Under the decomposition (1), the observed path is what $\rho > 0$ plus event-selection generates; the reversal is therefore *consistent with* the echo channel and does not independently identify overshooting. We accordingly downgrade the reversal from "finding" to "pattern consistent with the echo interpretation," and note that the next-day open-to-close coefficient is small and insignificant ($t = 0.45$), which in the earlier draft was read as evidence for overnight diffusion and here is simply the echo's asymmetry between the overnight and intraday windows on an underpowered sample.

6.4 Portfolio sorts, honestly reported

Sentiment-sorted quintile portfolios on the baseline sample earn −20.3, +18.9, +6.5, +35.0, and +69.6 bps from Q1 to Q5, a Q5−Q1 spread of 89.9 bps with a Welch $t = 1.24$ ($p = 0.22$). We state three facts the earlier draft stated less prominently: the sort is *not monotone* (Q2 exceeds Q3); the spread

is not statistically significant; and the IC and direction tests are null (mean daily IC +0.035, $t = 0.30$ over 25 usable cross-sections; direction logit $p = 0.20$, marginal effect 11.5 points). Under the audit's standard, the economic-significance evidence does not clear its own bar, and gross-of-cost spreads on an event-selected sample of 47 observations per quintile should not be quoted as tradable. No transaction-cost claim is made beyond the observation that any cost level above zero further erodes a statistically insignificant spread.

6.5 Heterogeneity: interactions, not splits

The earlier draft split the sample at medians and reported that predictability "concentrates" in large firms ($t = 2.51$ vs. -0.03), multi-headline days ($t = 2.28$ vs. 1.39), and high-volatility regimes ($t = 2.49$ vs. 1.00) —the split-sample rhetoric the econometrics literature warns against, since separate significances do not test a difference. Table 5 reports the interaction-term tests. The size interaction is significant ($t = 2.25$, $p = 0.025$): predictability genuinely concentrates in above-median-dollar-volume firms, which in this corpus is nearly mechanical —mega-caps dominate search coverage, so the instrument is most precise (densest text input) exactly there, the interpretation the earlier draft reached by a wrong route. The news-intensity interaction is marginal ($t = 1.90$, $p = 0.057$). The volatility interaction is zero ($t = 0.01$): the earlier draft's high-volatility claim, based on coefficients that differed by 0.0007 across regimes with different standard errors, does not survive a formal test, and we retract it. Figure 8 contrasts the two inference styles directly —the point of the exercise is methodological as much as substantive.

Table 5: Heterogeneity: formal interaction tests versus split-sample statistics

Channel	Interaction coefficient	t	p
Sentiment × large firm (above-median dollar volume)	0.0248	2.25	0.025
Sentiment × multi-headline firm-day	0.0512	1.90	0.057
Sentiment × high-volatility regime	0.0001	0.01	0.994

Interactions from $r_{i,t+1} \sim$ sentiment + sentiment × modifier + modifier + same-day return + controls, two-way clustered, $N = 205$. The volatility-regime split of the earlier draft ($t = 2.49$ vs. $t = 1.00$) implies an interaction indistinguishable from zero.

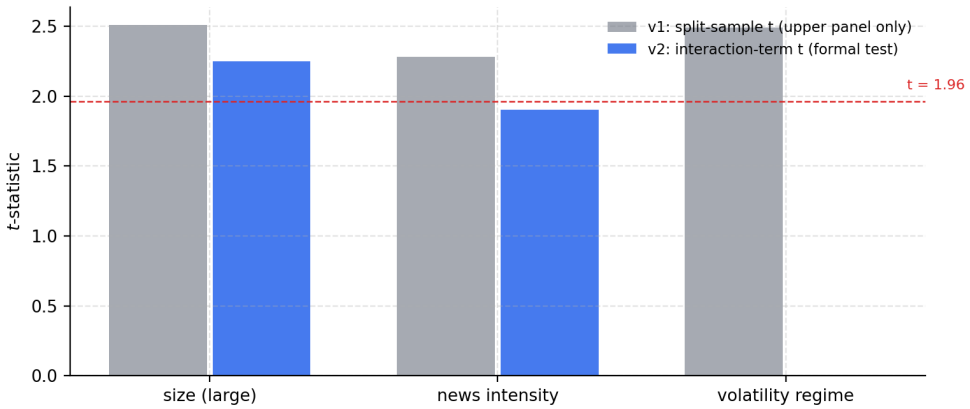


Figure 8: Split-sample t-statistics (grey) versus formal interaction t-statistics (blue) for the three heterogeneity claims of the earlier draft. Only the size effect survives formal testing; the volatility claim reverses to zero.

6.6 Threats to validity we could not address

Four threats remain outside the audit's reach. *Publication-timing granularity*: headlines carry dates but not timestamps, so the same-day return attributed to a news day may include pre-publication price movement, and we cannot separate before-open, intraday, and after-close news —the overnight/intraday split is a coarse proxy, and all timing claims built on it are exploratory. *Single scorer*: all conditions use one

model endpoint; cross-model stability (the model-variability problem of Herrera-Poyatos et al., 2025) is not tested here. The audit rounds ran twelve hours after the original round, and the identical named protocol drifted +5.8 points in level between sittings; the interleaved session has since decomposed that drift into a +3.6-point batch-composition component and a non-monotonic endpoint component of about ±1 point per session-pair (Section 5.2), so the drift is no longer unattributed —but its endpoint component remains outside our control, and the repeat-call ICC therefore bounds near-term stability only. *No human gold standard*: without human sentiment labels we can audit the score's relation to returns and to its own perturbations, but not its semantic accuracy against ground truth; the price-language classification audit of Section 5.4 is LLM-assisted and does not substitute for one. *External validity*: 36 mega-caps over 74 days in a single (calm-to-moderate) regime; nothing here transfers to small caps, other markets, or stress periods, and the identity level effect —about −3 points on this corpus —may differ where firm priors are stronger or weaker.

7 What Survives, and for Whom

7.1 The synthesis

Assembling the five audit results, with the confirmatory session added: the association between LLM news sentiment and next-day returns on this sample is (i) marginal under family-wise standards, weakening with the same-day control, and *draw-dependent in its nominal significance* (t between 1.47 and 2.12 across ten re-scorings of the identical headlines); (ii) indistinguishable, as a predictive coefficient, from the anonymized condition and from the dictionary in every contrast fielded, with its apparent identity level shift re-attributed first to between-session drift and then, by the interleaved experiment, to a genuine −3-point identity prior (named scores sit below anonymized twins) plus a +3.6-point batch-composition effect; (iii) measured by an instrument that is reliable but stochastic, with reliability itself stable across sittings (ICC 0.949 within each session); (iv) concentrated in headlines that describe realized price moves —a classification that survives its own audit —with the same-day co-movement immune to anonymization; and (v) estimated on a sample selected on realized activity. Under the decomposition (1), the reading is: ι real but small and negative, η mostly batch composition, $\mathrm{Var}(\varepsilon)$ substantial, ρ carrying most of the observed $\hat{\beta}$, and $\Pr(D \mid |r|)$ increasing. Nothing in this list refutes Lopez-Lira and Tang (2026), whose evidence rests on a commercial news archive, a much longer sample, and a within-window design whose power dwarfs ours. What the list does is bound what a *small-sample, search-collected, single-call* pipeline can claim: not a discovery of predictability, but a characterization of what such pipelines actually measure.

7.2 Implications for the LLM-finance literature

Three implications, stated as protocol rather than doctrine. First, *anonymize by default, verify the anonymization, and field the contrast interleaved in one session*: every LLM-sentiment study should report the named-versus-anonymized level shift and predictive difference on its own corpus. The experiment is cheap (one additional scoring pass), it separates priors from information (H2), and in post-cutoff windows it is the direct test of whether identity effects survive the removal of memorization as an explanation —complementing the in-sample results of Glasserman and Lin (2023). Two protocol details are now load-bearing because this audit got both wrong in sequence. Field the contrast in the same sitting as its named twin: the initial estimate here was confounded by a twelve-hour gap that carried a +5.8-point drift. And hold the batch composition fixed: when the repair experiment re-scored the identical headlines inside the original 542-article batches, levels moved +3.6 points for that reason alone —re-batching is itself a treatment. The audit also caught a de-identification failure in its own anonymizer (ten headlines still naming the focal firm; Appendix B), a reminder that the anonymization step needs a verification pass, not just a run. Second, *report the noise floor*: an ICC or repeat-call dispersion should accompany any single-call scoring result, because the attenuation it implies is a first-order input to coefficient magnitudes (Remark 1) and because "30% exact replication across four calls under default decoding" contradicts the intuition

most pipelines operate on. Third, *audit the text for echoes and the sample for selection*: search-collected and even wire-collected corpora contain retrospective coverage; a price-language flag is a five-line regular expression —and, as this revision shows, the flag itself should be spot-audited against a written rulebook, because 3 of 52 matches were retail promotions and 3 more were revenue language —and a coverage-selection test is a two-line fixed-effects regression. Both took minutes to compute and both materially changed this paper's conclusions.

The deeper point concerns inference culture. The earlier draft of this project committed, in sequence, the classic errors: inferring instrument dominance from separate significances (corrected in Table 3), inferring heterogeneity from split-sample significances (corrected in Table 5), characterizing a non-monotone sort as monotone, asserting sample neutrality that a two-sample test rejects, and interpreting a mechanically-generable horizon pattern as evidence for a behavioral mechanism. None of these errors required dishonesty; all of them required only the standard narrative pressure of turning marginal statistics into a story. AI-assisted pipelines amplify the risk —the marginal cost of generating a plausible table-and-narrative has collapsed —which is precisely why the audit layer (pre-specified families, difference tests, permutation and bootstrap inference, released artifacts) is now the quality-bearing layer of this kind of research.

7.3 What would change our conclusions

The audit's verdict is conditional on this sample, and three findings would overturn parts of it. A longer window with wire-quality timestamps that preserved the baseline coefficient on *clean* headlines would establish the semantic channel here absent ($\kappa > 0$ in the return equation, not just the score equation). A cross-model replication with the interleaved design that found a materially different identity shift would establish how firm-specific the -3-point prior is —the within-model estimate here is precise, but a single scorer bounds generalization. And a coverage-neutral corpus (fixed firm list, all headlines, no search ranking) that reproduced the predictive coefficient would eliminate the selection bound of H5. Each of these is a concrete, budgetable study; none is this paper.

8 Limitations

Power. With 205 regression observations, 35 firms, and 59 trading days, the audit cannot rule out predictive coefficients of economically meaningful size; the confidence intervals in Figure 2 are wide in both directions. Our null results on differences (H2) are statements of non-estimability at this sample size, not of equality; we report them as such, and the correct inference from Table 3 is that instrument comparisons require samples an order of magnitude larger.

Corpus. The headlines come from search-engine results at collection time, not a news archive: coverage is thin, selected on activity (H5), and enriched in retrospective and prediction-style articles. This is a limitation for generalizing the baseline result and an asset for the echo analysis; we use it as the latter while flagging the former.

Instrument. One model, one endpoint, default decoding, no version pinning beyond what the provider exposes. The four scoring conditions were fielded in two sittings twelve hours apart; the interleaved third session has since attributed most of the between-sitting drift (+5.8 points) to batch composition (+3.6) rather than endpoint change, and the endpoint residual is non-monotonic —but the provider could change the served model at any time without notice, so no long-horizon stability is certifiable, and cross-model generalization is untested. The anonymization is conservative (names and tickers only): product names and stylistic cues survive, and 5 of 341 anonymized headlines (1.5%) still name another sample firm —typically an analyst house or co-mentioned mega-cap (Appendix B) —so the anonymized condition remains a lower bound on de-identification.

Alignment. Publication dates without timestamps prevent event-window precision; same-day returns conflate pre- and post-publication movement; the overnight/intraday decomposition is coarse. Mechanism

claims about timing (attention diffusion versus liquidity) are therefore not identifiable in this design, which is why the paper makes none.

Production. The research pipeline, including this draft, was produced with AI assistance under human direction. We mitigated the attendant risks by pre-specifying the test family before running the augmented inference, recomputing every number in this draft from archived artifacts, and retracting specific claims of the earlier draft explicitly rather than silently. The replication package is the source of truth; where this text and the package disagree, the package is right.

9 Conclusion

This paper took a working result —LLM news sentiment predicts next-day stock returns, and beats dictionaries doing it —and asked the question the result's popularity has outrun: what is the score actually made of? On a frozen corpus of 341 headlines covering 36 U.S. mega-caps over 74 trading days, scored under four conditions that each vary one ingredient of the measurement and then re-scored in a pre-specified interleaved session, the audit returns a five-part answer. The baseline association is marginal, specification-sensitive, and draw-dependent: across ten re-scorings of the identical headlines the coefficient is stable but the t-statistic ranges over $[1.47, 2.12]$, so nominal significance depends on the day the pipeline scores ($t = 2.10$, BH $q = 0.062$, $t = 1.82$ with the same-day control on the published draw). A 4.75-point named-versus-anonymized gap that initially looked like a firm prior decomposes, under the same-session benchmark, into +5.79 points of between-session drift plus a −1.04 residual —and the interleaved experiment then settled both layers: the drift is itself mostly a +3.6-point *batch-composition* effect (pooled $z = 3.4$ across two sittings with inverted run order), and the identity prior it masked is real, negative, and about three points (−2.82; round-level $t = 6.8$, call-clustered $t = 2.7$; every round-level estimate negative). The instrument is reliable but stochastic (ICC 0.919 pooled, 0.949 within every sitting; 29.9% exact replication across four calls), its reliability is stable even while its levels move, and its noise decomposes into a response-level mood shared by each batch's 25 scores (13% of variance) plus idiosyncratic noise. The predictive loading concentrates in the 15% of headlines that describe realized price moves —a classification that survives its own audit and whose split no single firm or day carries —and the same-day co-movement of sentiment and returns survives anonymization, placing that channel in the text rather than the model's memory. And coverage itself is selected on realized activity, so the sample over-weights the events where non-semantic continuation and reversal are strongest.

The contribution is not a new predictor. It is a demonstration, on a real corpus with real returns, that the channels a measurement-validity literature has been studying separately —identity priors, sampling noise, text-level echoes, sample selection, and (newly quantified here) batch context —can be measured jointly and cheaply, and that doing so changes the interpretation of a headline statistic —in our own case, it caught a session-confounded conclusion that the first audit pass had reported as a finding, and then caught the confound's own confound when the repair experiment attributed most of the "session drift" to batch composition. The protocol that fell out of this exercise is deliberately minimal: anonymize once, verify the anonymization, and field the contrast interleaved in one sitting; hold the batch composition fixed across anything you compare; repeat-call once and report the noise floor; flag the text for realized-price language and audit the flag against a written rulebook; test the sample for selection; and pre-register the family before looking at returns. Every component is implementable in an afternoon at the marginal cost of one additional scoring pass —and the interleaved re-fielding, the one component that costs a full session, is the one this paper needed to correct its own sign.

Two conclusions for the literature. For studies that use LLM sentiment as a return predictor: a significant coefficient on a single-call, named, search-collected corpus is not evidence about text semantics until the echo and selection channels are ruled out; the burden of proof is now demonstrably cheap to meet. For the emerging audit literature: the post-cutoff window is the clean laboratory for identity effects, and two signatures documented here —the mild negative identity prior (−3 points, interleaved) and the batch-composition effect (+3.6 points for re-batching alone, robust to inverted run order) —are precise, cheap targets for cross-model and cross-market replication, with the open question of whether the composition

effect's asymmetry generalizes. The questions worth asking have shifted from whether LLMs can read the news to what, exactly, they are reading —and how much of the market, the firm, and the neighboring headlines is in the score before the text ever arrives.

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A LLM Scoring Prompt

The following system prompt was applied verbatim to every batch of 25 headlines in all four conditions; the named and repeat conditions prefix each headline with "[Company]", and the anonymized condition presents the de-identified headline text with no prefix. The model was accessed through a fixed API configuration with default decoding parameters; batches that failed to parse were retried up to three times. All raw responses and batch logs are archived in the replication package.

```
You are an expert financial analyst specialized in news sentiment analysis for
equity research. You evaluate how news headlines affect a company's stock value.

For each headline you must return:

- "sentiment": integer from -100 to +100. -100 = extremely negative impact on the
company's stock value; 0 = neutral or no clear impact; +100 = extremely positive
impact. Judge from the perspective of a shareholder of the company.

- "relevance": integer from 0 to 100. 100 = headline is directly about the company's
business, earnings, products, guidance, or stock; 0 = headline mentions the company
only incidentally or is a generic market/quote page.

Rules:

- Base your judgment ONLY on the information in the headline text.

- Earnings beats, raised guidance, product launches, upgrades, buybacks = positive.

- Earnings misses, cut guidance, lawsuits, recalls, downgrades, layoffs = negative.

- Speculation or rumor framing lowers the magnitude toward zero.

- Generic quote pages, price summaries without direction, or calendar pages =
sentiment 0, relevance low.

Return ONLY a valid JSON array. One object per headline, in the same order as given,
with fields: "id", "sentiment", "relevance". No other text.
```

B Anonymization Procedure

Company identifiers are removed by a case-insensitive alias table mapping each firm's name, common short forms, and ticker to the phrase "the company" (e.g., "Apple Is Up 20% in 2026. What Will It Take for AAPL Stock" becomes "the company Is Up 20% in 2026. What Will It Take for the company Stock"). 325 of 341 in-window headlines (95.3%) are altered. The procedure is conservative: product names, third-party firms mentioned inside a headline, and stylistic cues are left intact, so the anonymized condition is a lower bound on de-identification. Quantifying the residual firm-identity component under the executed alias table: 5 of 341 anonymized headlines (1.5%) still name at least one other sample firm —under both a case-insensitive and a case-sensitive matching convention —with analyst actions ("BofA," "Morgan Stanley") and co-mentioned mega-caps the typical cases. The dominant de-identification gap is therefore product language (e.g., device or franchise names) rather than residual firm names. Ambiguous tickers that collide with common words (e.g., "COST," "CAT," "V") are replaced only in their exact all-caps form.

A disclosed failure of the procedure. The executed alias table contains no entry for ticker V (whose identifiers would have been "Visa"), and all ten in-window V headlines contain the word "Visa" but never the standalone all-caps token "V." The anonymization was therefore a no-op for those ten headlines: the text sent to the model under the anonymized condition still named the focal company —2.9% of the 341 —and the condition contrast for them is effectively named-versus-named. Five of the ten are U.S. State Department "Visa Bulletin" immigration headlines (unrelated to the payments company; all scored at relevance 20, so they never enter the firm-day sample but do enter the article-level tests), and five enter the relevance-qualified sample. The failure is instructive in two ways. First, it materially moves the marginal same-session identity contrast: excluding the ten V headlines sharpens it from -1.04 ($p = 0.11$)

Table 6: News coverage by firm (in-window articles and trading-day firm-days)

Ticker	Articles	Firm-days	Ticker	Articles	Firm-days	Ticker	Articles	Firm-days
AAPL	16	9	GE	8	6	MSFT	10	5
AMZN	8	6	IBM	8	7	NFLX	14	11
BAC	7	5	JPM	4	4	NVDA	13	8
CRM	17	8	MA	9	8	ORCL	12	9
GE	8	6	MSFT	10	5	PFE	5	4
IBM	8	7	ORCL	12	9	PG	6	4
JPM	4	4	QCOM	7	7	QCOM	7	7
MA	9	8	WFC	12	9	TSLA	9	8
MSFT	10	5	—	—	—	V	10	3
ORCL	12	9	—	—	—	WFC	12	9
QCOM	7	7	—	—	—	WMT	13	9
WFC	12	9	—	—	—	XOM	8	7

Articles: headlines dated 2026-07-01 to 2026-10-14 in the archived corpus (column total 341). Firm-days: unique firm–date cells with scored relevance ≥ 30 falling on a trading date (column total 240, equal to the regression sample). 36 firm–date cells fall on non-trading dates and are dropped without roll-forward (Section 4). Computed by `coverage_table.py` from `news_all.json`, `llm_scores.json`, and `panel.csv`.

to -1.29 ($p = 0.04$; Wilcoxon $p = 0.03$), which is why Section 5.2 reports the contrast as a range rather than a verdict. Second, and more generally, it was caught only by re-running the anonymizer with a verification pass —checking that anonymized text no longer matches the focal firm's identifiers —which we now treat as a mandatory step of the protocol rather than an optional one. The illustrative pair: the named condition "[Visa] Visa Stock Just Posted Its Best Quarter Since 2022." scored +80 (relevance 90), while the "anonymized" condition —the identical text minus the prefix —scored 0; the difference is a clean single-headline demonstration of how much identity information the prefix-plus-name carries. Computed by `v_exclusion.py` from the archived corpus and scores.

C Firm-Level Coverage

Table 6 reports news coverage by firm: in-window headlines and relevance-qualified trading-day firm-days. Coverage ranges from 4 to 17 articles per firm and correlates with firm prominence in web search, which motivates the selection analysis of Section 5.5 and the size interaction of Table 5. The table is generated directly by `coverage_table.py` from the archived corpus, scoring, and panel files, so every cell is reproducible; the earlier hand-assembled version of this table (37 tickers including a firm with no archived records, inconsistent column totals) has been withdrawn.

D Archived Results from the Earlier Draft

For transparency, Table 7 reproduces the horizon and portfolio statistics of the earlier draft exactly as archived, with the audit's re-interpretation noted alongside. These numbers are reproducible from the archived v1 analysis script and are unchanged by the audit; what changed is their interpretation.

E Power and Minimal Detectable Effects

For each test in the declared family, Table 8 reports the observed effect, its standard error, the two-sided 5% minimum detectable effect at 80% power ($\text{MDE}_{80} = (z_{0.975} + z_{0.80}) \times \text{SE} = 2.80 \times \text{SE}$), and the power of the test against the observed effect, $\Phi(|\delta|/\text{SE} - 1.96) + \Phi(-|\delta|/\text{SE} - 1.96)$. Standard errors for regression coefficients are the two-way clustered values of Table 2; standard errors for the difference tests are implied by the bootstrap confidence intervals of Table 3 (width divided by $2{\times}1.96$); the level-shift rows

Table 7: Earlier-draft statistics, reproduced and re-interpreted

Statistic	Value	Earlier interpretation	Audit interpretation
Coefficient, $t+2$ horizon	-0.0080 ($t = -2.04$)	attention-driven reversal	echo+event-selection dynamics
Coefficient, $t+3$ horizon	-0.0051 ($t = -1.52$)	reversal decay	same, underpowered
Next-day intraday coefficient	0.0018 ($t = 0.45$)	overnight diffusion	echo asymmetry, noisy
Q5−Q1 spread	89.9 bps ($t = 1.24$)	economic significance	insignificant; non-monotone
Mean daily IC	0.035 ($t = 0.30$)	weakly consistent	null
Direction logit	$p = 0.20$	power-limited	null
High-vol. vs. low-vol. coefficients	0.0135 vs. 0.0128	state-dependence	interaction $t = 0.01$: retracted

use their paired standard errors; the selection row uses the two-way clustered standard error of the fixed-effects specification of Section 5.5. The table quantifies what the paper's non-rejections can and cannot rule out: the named-versus-anonymized predictive difference rules out substitutions larger than 0.006 — about 42% of the baseline coefficient itself —while the named-versus-dictionary contrast rules out almost nothing ($\text{MDE}_{80} = 0.094$, more than six times the baseline). At the return layer, the design can only detect next-day coefficients of 0.020 or larger (72 bps per one-standard-deviation sentiment move, against the observed 54 bps), and the achieved power against the observed effect is 0.56: the marginal significance of H1 is exactly what an underpowered design produces around a true effect of this size. The H4 echo split has power 0.31 against its observed difference, which is why Section 5.4 labels it directional rather than established. At the measurement layer, the level-shift and selection tests are comfortably powered (0.999 and 0.79), which is the statistical counterpart of the paper's narrative: the channels the audit finds with confidence are the ones it can measure. The same-session identity row is the revision-stage contrast: at $|\delta| = 1.04$ points and SE = 0.65, the design reaches power 0.36 —the marginal mean-test result is what a design of this size produces around a one-point effect, and the rank test and V exclusion are the sharper instruments (Section 5.2).

Table 8: Power and minimal detectable effects for the declared family

Test	Observed	SE	MDE_{80}	Power
H1: baseline next-day coefficient (named)	0.0149	0.0071	0.020	0.56
H2: $\beta_{\text{named}} - \beta_{\text{anonymized}}$	0.0004	0.0023	0.006	0.05
H2: $\beta_{\text{named}} - \beta_{\text{dictionary}}$	−0.0119	0.0335	0.094	0.07
H4: $\beta_{\text{price-lang.}} - \beta_{\text{clean}}$	0.0436	0.0297	0.083	0.31
H2: identity level shift (points, $n = 341$)	4.75	0.92	2.59	1.00
H5: coverage selection (bps, $n = 2{,}628$)	61.1	22.2	62.1	0.79
H2: same-session identity shift (points, $n = 341$)	−1.04	0.65	1.83	0.36
H2: interleaved identity shift (points, $n = 341$, session 3)	−2.82	1.03	2.88	0.78

Two-sided 5% tests; $\text{MDE}_{80} = 2.80 \times \text{SE}$. "Power" is achieved power at the observed effect. The identity-level-shift row is the cross-session contrast as declared; the same-session row is its revision-stage re-estimate; the last row is the interleaved-session confirmatory contrast under the conservative batch-call-clustered SE (the headline-level SE of 0.66 would give power 0.99; the round-level meta-test is stronger still). The H4 and H5 rows use the $B = 9{,}999$ bootstrap CI and the fixed-effects clustered SE of the v4 inference upgrades. Computed by `revision_power.py` (family and session-2 rows) and `analysis_v3.py` / `reinference.py` (session-3 row) from archived artifacts.

F Session Composition and Scoring-Session Drift

Table 9 documents the three scoring sittings. The original named round scored all 542 headlines in two runs on 2026-10-15 (16:01 and 16:24 UTC); the audit round fielded the anonymized condition and three named repeats on the identical 341 in-window headlines in a single sitting on 2026-10-16 (04:24–04:33 UTC); the confirmatory session interleaved four named and four anonymized rounds plus the 542-composition control on 2026-10-18 (04:54–05:34 UTC, from the millisecond log). The named protocol is verbatim-identical across sittings (system prompt, user-message format, batch size 25, decoding disabled thinking); the v1 batches additionally contained the 201 out-of-window headlines, so batch composition differs between sittings —and Appendix G measures exactly how much that difference is worth. A provenance distinction is recorded for the timestamps: the session-2 and session-3 times are read from millisecond-resolution entries in the batch logs (`llm_score_v2.log`, `llm_score_v3.log`), while the v1 scorer logged progress without timestamps, so the session-1 times are the modification timestamps of the first run's log file and of the archived output file (16:01:25 and 16:24:00 UTC respectively) —they mark the end of each run rather than its start, and are file-system evidence rather than in-log evidence; both record types are archived.

Table 9: Scoring sessions and cross-session stability of the named protocol

Quantity	Value
Session 1 (named, 542 headlines)	2026-10-15, 16:01–16:24 UTC
Session 2 (anonymized + 3 named repeats, 341 headlines)	2026-10-16, 04:24–04:33 UTC
Session 3 (4 named + 4 anonymized interleaved, + 542-composition control)	2026-10-18, 04:54–05:34 UTC
Gap, session 1 to session 2	$\approx$ 12 hours
Named level drift, session 1 − session 2 (points)	+5.79 (t = 7.03, $p < 10^{-11}$)
Same-session identity shift, named − anonymized (points)	−1.04 (t = 1.60, p = 0.111)
excluding the ten ticker-V headlines	−1.29 (t = −2.05, p = 0.041)
Identity shift, interleaved session 3 (points)	−2.82 (round-level t = 6.8; clustered t = 2.7)
excluding the ten ticker-V headlines	−3.13 (clustered t = 4.8)
Batch-composition effect, within session 3 (points)	+3.61 (clustered t = 2.7)
Session 4 replication, run order inverted (points)	+3.50 (clustered t = 2.2)
Decomposition check	+5.79 − 1.04 = +4.75 (published shift)
Mean cross-session pairwise r (original vs. repeats)	0.905
Mean within-session pairwise r (repeats vs. repeats)	0.950
Mean |diff|: cross-session / within-session (points)	11.3 / 6.5
Anonymized-vs-repeat / repeat-vs-repeat |diff| (points)	7.9 / 6.5
ICC(2,3), named within session 2	0.949
ICC(2,4), named within session 3	0.949
ICC(2,4), pooled across sessions (published)	0.919

Article level, n = 341. Cross-session agreement is significantly worse than within-session agreement (Mann–Whitney on |diff| distributions, $p < 10^{-6}$): the scoring process varies between sittings by more than call-level noise alone. What looked like endpoint drift is now largely attributed: the batch-composition row shows that +3.61 of the +5.79-point cross-session gap is reproduced by composition alone inside a single sitting (Appendix G), with a non-monotonic endpoint residual of ±1 point per session pair. Rows computed by `revision_power.py` and `v_exclusion.py`; session-3 rows by `analysis_v3.py`; raw batch logs in `scripts/llm_score*.log`.

Three interpretive cautions. First, the drift is a level phenomenon: session 1's named scores are six points more bullish on average than session 2's named scores of the identical headlines, while the slope-relevant structure (which headlines are relatively positive) is largely preserved (r = 0.905 across sessions). Second, the published ICC(2,4)= 0.919 pools draws from both sittings and therefore already incorporates the cross-session component; within-session reliability is higher (0.949), and the interleaved session shows that within-session reliability is itself stable across sittings —0.949 in both. Third, the direction of endpoint drift is not a constant: the third sitting did drift back toward session-1 levels (+1.14 points at the 341 composition), exactly the caution this appendix originally raised. The honest constraint on pipelines remains a design rule rather than a correction: contrast conditions within a sitting, freeze the batch composition, and treat any cross-sitting comparison as confounded by construction.

G The Interleaved Replication Session

The confirmatory experiment was pre-specified in the revision problem list (P1-2/P1-3) before any session-3 data existed: re-field named and anonymized conditions on the identical 341 headlines *in one sitting*, with condition order alternated at the batch level, at least four rounds per condition, plus one pass of the identical headlines inside the original 542-article batch structure as a composition control. The executed design: four rounds; within each round, fourteen batches of 25; even-numbered batches call named-then-anonymized and odd-numbered call anonymized-then-named; eight score files plus the composition file; every call logged with a timestamp and model identifier (`glm-4-plus`); the analysis is `analysis_v3.py` and the raw log is `scripts/llm_score_v3.log`. Batch parity is a fixed property of a headline (its position in the stream), so the parity contrast identifies call-order contamination, not order effects: the identity effect is 3.34 points (anonymized-minus-named) on named-first batches versus 2.27 on anon-first batches —no artifact.

The composition decomposition is exact by transitivity, and each leg is separately interpretable:

Table 10: Decomposing the cross-session named-level drift (points, article level, $n = 341$)

Cell	Sitting	Mean score
Named, 542-article composition	2026-10-15 (session 1)	+16.88
Named, 341-only composition	2026-10-16 (session 2)	+11.09
Named, 341-only composition	2026-10-18 (session 3)	+12.22
Named, 542-article composition	2026-10-18 (session 3)	+15.84
Legs of the decomposition		
Endpoint, session 1 − session 3 (both 542-composition)		+1.05
Composition, 542-composition − 341-composition (within session 3)		+3.61 ($t = 4.55$)
Endpoint, session 3 − session 2 (both 341-composition)		+1.14
Total, session 1 (542) − session 2 (341)		+5.79

The four cells identify the additive decomposition exactly: total drift = endpoint leg + composition leg + endpoint leg. The composition leg is measured inside a single sitting —same endpoint, same protocol, only the surrounding batch content differs — so re-batching alone accounts for 62% of what the two-session evidence had attributed to "the session." The endpoint legs are non-monotonic (session 3 sits between sessions 1 and 2), so the endpoint residual is a ±1-point-per-session-pair wander, not a trend. Computed by `analysis_v3.py` from `llm_scores_s3_*.json` and the archived session-1/2 scores.

Anonymized levels track the same structure: +12.13 (session 2) and +15.04 (session 3, four-round average) —the anonymized protocol moves between sittings too, which is why cross-session named-versus-anonymized contrasts absorb drift on *both* sides.

The mechanism sitting (2026-10-18, fourth session). A fourth sitting ran the discriminating design the problem list pre-specified, with the run order *inverted* so that time-of-session works against the composition effect: (i) the 542-article composition pass ran first, (ii) the standard 341 pass second, and (iii) a seeded random shuffle of the 341 third (seed 20261018, order archived as `s4_shuffle_order.json`; the original order is company-blocked by collection, the shuffle mixes companies within every batch). Three results. First, the composition effect replicates under the inverted order at +3.50 (batch-call-clustered $t = 2.17$, $p = 0.042$; pooled with session 3: +3.56, $z = 3.4$): since the 542 pass ran *early* this time, a time-of-session explanation would have flipped the sign —the effect is batch composition, not sitting position, and the drift between sessions 3 and 4 at fixed composition is +0.57 ($t = 0.3$). Second, re-batching the *same* 341 headlines at random moves levels by +4.72 (clustered $t = 2.02$, $p = 0.065$ —suggestive, one sitting): batch assignment matters even within a fixed corpus, so the composition channel is not specifically about out-of-window headlines. Third, the mechanism:

The mechanism table carries the paper's most cautionary arithmetic. A naive reading —"anchoring $\theta = 0.36$, batch means differ, therefore levels shift" —fails on the telescoping identity: linear anchoring redistributes scores *within* a corpus but provably cannot move the average, so the observed +3.5-point shifts must live in the non-linear part (the estimated asymmetry) or in interactions with response mood. The practical rule is unaffected by the open derivation: hold the batch composition fixed across anything

Table 11: Mechanism statistics (eight named-condition passes, sessions 3–4)

Quantity	Estimate
Clean anchoring on batch-mates' fixed content, θ	+0.36 (t = 4.9)
asymmetry: extra slope when batch-mates' mean is positive	+0.69 (t = 2.3; negative-mate slope ≈ 0)
Response-level mood (shared by the 25 scores of one API response)	SD 2.6; 13% of residual variance
Idiosyncratic per-headline noise	SD 6.7; 87%
Level shift predicted by *linear* anchoring under any re-batching	exactly 0 (telescoping identity)
Observed level shifts: 542-composition / random re-batch	+3.56 pooled (z = 3.4) / +4.72 (p = 0.065)

The anchoring coefficient regresses each headline's score on the mean of its batch-mates' grand means (estimated across all passes, noise-averaged), with headline and pass fixed effects —the regressor is a deterministic function of the batch assignment, free of same-response noise endogeneity. The telescoping identity: for any partition into balanced batches, the average of *leave-one-out batch-mate means* over all headlines equals the global mean, so a linear anchoring term cannot move the corpus mean; observed level shifts therefore require the estimated non-linearity (asymmetric assimilation: positive batch-mates pull scores up, negative ones do not push them down) and/or interactions with the response-level mood. A full quantitative account of the shifts from these coefficients alone is not closed —honestly: the *existence* of batch effects is established at $z \approx 3$–5 across two sessions with inverted run order; the *mechanism* is pinned to the response (anchoring is real, asymmetric, and level-neutral in its linear part) but not fully derived. Computed by `anchoring_regression3.py` and `reinference.py`.

being compared, because re-batching moves levels by more than most effects of interest, and the direction of the move is not predictable from the batch means alone.

H Classification Audit of the Price-Language Flag

All 52 headlines flagged by the realized-price regular expression of Section 5.4 were re-classified one by one against a written rulebook with four categories: **P** —the headline states or clearly implies the company's stock price has already moved (including moves stated with a fundamental cause, e.g. "shares jump on blowout results"); **E** —the matched keyword is a fundamental magnitude ("revenue rises 12%"), with no realized stock move; **D** —a directional forecast only ("could sink," "may drop to $300"); **N** —a regex false positive ("execution short fall"; "Walmart just dropped October deals" —retail promotions). The tie rule: dual-content headlines with both a realized move and a fundamental result classify as P, since the defining feature of an echo headline is that the move is reported. The classifier is disclosed for what it is: LLM-assisted, structured, item-by-item, run by the same class of model that produced the scores —an internal consistency audit of the flag, not independent human annotation, and not a gold standard (the rulebook and all 52 labels are archived as `priced_classification.json`).

Table 12: Auditing the echo classifier and the echo split

Quantity	Value
Flagged headlines / re-classified as realized-price (P)	52 / 43 (82.7%)
Fundamental magnitude (E) / directional forecast (D) / false positive (N)	3 / 3 / 3
Firm-days, regex flag vs. strict-P	38 vs. 33 (20 companies)
β, regex-flagged vs. strict-P (same-day control spec)	0.0470 (t = 1.80) vs. 0.0475 (t = 1.61)
Mean |same-day return|, strict-P vs. clean vs. flagged-non-P	361 / 178 / 120 bps
Largest single-firm share of strict-P firm-days	9%
Leave-one-firm-day-out: β range / sign flips of diff	$[0.018, 0.059]$ / 0
Leave-one-company-out: β range / sign flips of diff	$[0.019, 0.056]$ / 0

The refined split uses only firm-days with at least one strict-P headline. The misclassified firm-days' 120 bps of same-day absolute return —below even the clean group's 178 —is the strongest internal validation of the classification: headlines that are not echoes do not behave like echoes. Jackknife rows drop each strict-P firm-day (or each company holding one) and re-estimate both split coefficients; the difference never changes sign in any replicate. Computed by `p11_classification_analysis.py` from `priced_headlines.json`, `priced_classification.json`, and the archived panel.